

FINITE-USE GEOMETRY AND LEARNING OF ORDERED BINARY QUANTUM MEASUREMENTS

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ABSTRACT. We study the full body of ordered binary quantum measurements with classical output and no residual quantum output. A regularized Sylvester form at the midpoint of two effects is comparable, with explicit dimension-dependent constants, to their finite-use adaptive distance, including arbitrary references and every spectral rank. In each fixed input dimension d , the small-error covering number has order $N^{d^2/2}[\log(N+2)]^{\lfloor d/2 \rfloor} \delta^{-d^2}$. Uniform volume bounds for actual operational balls and a nested endpoint integral account for the logarithmic multiplicity. We construct a common learner on the entire matrix body with $Cd^4N\delta^{-2}\log(d/\eta)$ training calls for error δ in the future- N adaptive distance and failure probability at most η . This learning guarantee holds for all $d, N \geq 1$, sufficiently small absolute $\delta > 0$, and $0 < \eta \leq 1/8$; each entangled block has at most N calls. A scalar converse makes the dependence on horizon, error, and confidence optimal for every fixed d . Simultaneous endpoint calibration controls the variance introduced by two-dimensional compressions, where a noise-adapted qubit learner applies. A deterministic finite rational construction attains the covering order, with exhaustive computation accounted for separately from the reusable payload. All boundary cases are included.

1. INTRODUCTION

Repeated observations impose different resolutions on different perturbations of a quantum measurement. An interior change of an eigenvalue is resolved on the scale $N^{-1/2}$, whereas a rotation of a projective measurement can be resolved on the scale N^{-1} . The two scales meet at singular effects and repeated eigenvalues. We determine the resulting geometry on a compact matrix body, calculate its description entropy, and construct a common learning procedure whose loss is the distinguishability of all future experiments with at most N uses.

Fix a finite input dimension d and consider

$$\mathfrak{E}_d = \{E = E^* \in M_d(\mathbb{C}) : 0 \preceq E \preceq I_d\}.$$

The effect E specifies the memoryless ordered binary measurement

$$\mathcal{M}_E(\rho) = \text{tr}(E\rho)|1\rangle\langle 1| + \text{tr}((I_d - E)\rho)|0\rangle\langle 0|.$$

The input is consumed and the output is classical. The distance d_N is the supremum of the unhalved trace norm between the final states of a common experiment with at most N calls. The experiment may entangle its inputs, retain an arbitrary reference and quantum memory, use feedback, and stop publicly. Thus $0 \leq d_N \leq 2$; classical total variation is one half of the corresponding classical trace norm. The nonadaptive restriction d_N^{na} retains entangled input blocks and references and omits feedback between calls.

1.1. An intrinsic comparison and its entropy. For $E, F \in \mathfrak{E}_d$ put

$$M = \frac{1}{2}(E + F), \quad H = F - E, \quad V = M(I_d - M),$$

and define the midpoint modulus

$$Q_N(E, F)^2 = N \langle H, (L_V + R_V + N^{-1} \text{Id})^{-1} H \rangle_{\text{HS}}.$$

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The inverse acts on the real Hilbert space of Hermitian matrices. Theorem 2.1 proves, for every $d, N \geq 1$,

$$(1.1) \quad \frac{\min\{1, Q_N(E, F)\}}{8192d} \leq d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F) \leq d_N(\mathcal{M}_E, \mathcal{M}_F) \leq \min\{2, 8Q_N(E, F)\}.$$

No spectral margin or multiplicity assumption is imposed. In particular $d_N \leq 65536d d_N^{\text{na}}$. This is a fixed-dimensional comparison of optimal values. The modulus Q_N is neither identified with an optimal value nor assumed to satisfy a triangle inequality.

A horizontal measurement dilation yields the adaptive tangent bound. A regularized Sylvester equation separates an arbitrary tangent from a remainder controlled by the hybrid inequality. Operator concavity of $G(I_d - G)$ then integrates the tangent bound along a finite segment, including boundary endpoints. The converse uses Bernoulli inputs and exact two-dimensional compressions. All required qubit statements and their proofs are included below.

Let $\mathcal{C}_N(\mathfrak{E}_d, \delta)$ denote the least number of legal memoryless binary centres needed to cover the body in d_N . For every fixed d , Theorem 3.1 gives

$$(1.2) \quad \mathcal{C}_N(\mathfrak{E}_d, \delta) \asymp_d N^{d^2/2} [\log(N+2)]^{\lfloor d/2 \rfloor} \delta^{-d^2}, \quad N \geq 1, \quad 0 < \delta \leq \delta_d.$$

The positive cap δ_d and the comparison constants may depend on d . The proof first bounds the weighted volume of actual operational balls uniformly over every boundary stratum. The total volume is then a spectral integral whose ordered endpoint sectors have cumulative exponents $S_j^2/2$, where S_j is a signed prefix sum. Balanced prefixes and matching nested endpoint boxes produce the logarithmic multiplicity $\lfloor d/2 \rfloor$.

1.2. Learning with a future operational loss. The statistical problem supplies calls to an unknown effect, without its matrix entries. For every fixed d , Theorem 5.1 constructs one learner, common to all $E \in \mathfrak{E}_d$, satisfying

$$(1.3) \quad \sup_E \mathbb{P}_E\{d_N(\mathcal{M}_E, \mathcal{M}_{\hat{E}}) > \delta\} \leq \eta, \quad M \leq C_d N \delta^{-2} \log(1/\eta).$$

Here $N \geq 1$, $0 < \delta \leq \delta_0$ for an absolute $\delta_0 > 0$, and $0 < \eta \leq 1/8$. The bound on the training calls M holds on every record. Each entangled block uses at most N calls. A scalar Bernoulli subfamily gives the matching lower order even for arbitrary reference-assisted adaptive training. More precisely, the upper bound is $Cd^4 N \delta^{-2} \log(d/\eta)$ with an absolute C . The dependence on d is not asserted to be optimal.

The construction begins with a qubit learner for unknown bias, contrast, and direction. Independent fair signs remove the bias from GHZ parity observations. A dyadic amplitude search stops at its first rejected scale, preserving a small-cone invariant; fresh observations complete the direction and endpoint estimates. Its confidence ledger controls both exceptional records and the maximum training cost.

For a general matrix effect, an additional calibration stage is essential. It estimates the two support boundaries in relative Loewner order at scale N^{-1} and controls the absolute matrix error at scale $N^{-1/2}$. In the calibrated eigenbasis, these two guarantees control the extra variance introduced by every two-dimensional compression. Qubit learning on those compressions then gives compatible matrix coordinates in the regularized quadratic form. Projection onto the legal effect body completes the estimate. The calibration and compression budgets are summed conditionally on the observed record, with no additional $\log N$ factor.

1.3. Deterministic descriptions and resources. Theorem 6.1 gives a deterministic finite rational codebook at the order in (1.2). Its greedy rule uses exact comparisons of Q_N^2 on a finite legal matrix grid. The operational separation in (1.1) controls its cardinality, and an inward rounding map gives coverage at every rank. The encoder and decoder reproduce the same

dictionary from the public parameters, with no supplied custom dictionary. Every fixed-length word decodes to a legal effect. The required payload is

$$\frac{d^2}{2} \log_2 N + \lfloor d/2 \rfloor \log_2 \log(N+2) + d^2 \log_2(1/\delta) + O_d(1).$$

The construction is finite and exact; its enumeration cost is large. Optimal payload length does not imply efficient classical computation. The direct qubit code remains available with its own detailed resource accounting. Corollary 6.2 combines the common learner with this exact encoder, attaining the learning and payload orders simultaneously without additional calls to the unknown device.

1.4. Prior work and organization. Horizontal Kraus gauges, channel-extension metrology, and integration of local distinguishability have established antecedents [22, 7, 23, 24, 25, 26]. Inverse-Sylvester geometry and spectral matrix integration are likewise classical [27, 28]. The explicit closed-body midpoint comparison, its nested endpoint entropy, and learning under the future operational loss are the theorem objects studied here.

Mele–Bittel [5] and Zambrano–Ramos-Calderer–Kuang [29] already provide common estimators for unknown measurements in one-use losses. Their access, rank, dimension, and confidence statements are compared at theorem level in Section 9. The distinction in (1.3) is the dependence on the future horizon under the anisotropic loss d_N . The phase-estimation components have antecedents in [31, 32]; the conditional construction used here is proved directly.

The matrix comparison and entropy are followed by the qubit learning construction, its extension to fixed matrix dimension, and the exact codebook theorem. The later qubit geometry and the final prerequisite appendix make the focused article self-contained. The complete research edition preserves the earlier instrument, preparation, coherent, readout, stochastic-realization, arithmetic, and causal developments with their original hypotheses and proofs. The structural companion treats repeatable nondisturbing classical observations and retains that separate physical interface.

2. MATRIX GEOMETRY IN ARBITRARY INPUT DIMENSION

The binary effect body has a finite-use description that does not require eigenvector coordinates. Its regularization is the same at every spectral rank, and two-dimensional compressions supply lower witnesses for the full matrix problem. Throughout this section $d, N \geq 1$ are integers and

$$(2.1) \quad \mathfrak{E}_d = \{E \in \text{End}(\mathbb{C}^d) : E = E^*, 0 \preceq E \preceq I_d\}.$$

The measurement $\mathcal{M}_E$ has the ordered effects $E, I_d - E$ and no residual quantum output. The distance d_N has the same reference-assisted adaptive meaning as in Theorem 7.1. Write d_N^{na} for its restriction to nonadaptive experiments, allowing entanglement across the calls and a retained reference. These experiments may depend on the known pair of hypotheses.

Put $\tau = N^{-1}$ and $V_E = E(I_d - E)$. On the real Hilbert space of Hermitian matrices with Hilbert–Schmidt inner product, define

$$(2.2) \quad \mathcal{A}_E(H) = V_E H + H V_E + \tau H,$$

$$(2.3) \quad g_{N,E}(H)^2 = N \langle H, \mathcal{A}_E^{-1}(H) \rangle_{\text{HS}}.$$

The operator $\mathcal{A}_E$ is strictly positive, including at a singular effect. For a pair E, F set

$$(2.4) \quad M = \frac{1}{2}(E + F), \quad H = F - E, \quad Q_N(E, F) = g_{N,M}(H).$$

In any orthonormal eigenbasis of M , with eigenvalues m_i and $v_i = m_i(1 - m_i)$, this is

$$(2.5) \quad Q_N(E, F)^2 = N \sum_{i,j=1}^d \frac{|H_{ij}|^2}{v_i + v_j + N^{-1}}.$$

Definition (2.4) makes the expression independent of all choices at repeated eigenvalues. We use it as a comparison modulus; a triangle inequality for Q_N is not required.

The inverse Sylvester operator is familiar from the Bures metric g^B on the positive definite cone [27]. Its normalization gives the algebraic identity

$$g_{N,E}(H)^2 = 2N g_{V_E + \tau I_d/2}^B(H, H).$$

Here H remains the effect tangent; the differential of the variance map $E \mapsto V_E$ is $H - EH - HE$.

Theorem 2.1 (Finite-use geometry of the matrix effect body). *For every $E, F \in \mathfrak{E}_d$,*

$$(2.6) \quad \frac{1}{8192d} \min\{1, Q_N(E, F)\} \leq d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F) \leq d_N(\mathcal{M}_E, \mathcal{M}_F) \leq \min\{2, 8Q_N(E, F)\}.$$

The constants are uniform in the horizon and in both effects, without a spectral margin or a condition on eigenvalue multiplicities. The adaptive upper bound permits arbitrary retained references, common feedback, and bounded public stopping. Each lower witness uses inputs in one fixed subspace of dimension at most two. Moreover,

$$(2.7) \quad d_1(\mathcal{M}_E, \mathcal{M}_F) = 2\|E - F\|_{\text{op}}.$$

The theorem includes every rank of projection and all of the intermediate singular strata. The variance $E(I_d - E)$, rather than the smallest eigenvalue of either outcome, determines the matrix weights. The factor N^{-1} makes the formula finite on the entire closed body. The proof first establishes a path inequality valid for every common adaptive tester, and then obtains the converse from one- and two-dimensional experiments.

2.1. A horizontal measurement dilation. Kraus-gauge bounds for channel estimation were developed in [22, 7, 23]. Their integration along channel paths appears in [25, Section III, equation (26)] and, with general adaptive bounds, in [26, Section IV.A]. We use an explicit horizontal lift for binary effects and control the finite-horizon regularization by a trace-norm remainder.

Lemma 2.2 (Horizontal derivative and regularized path bound). *Every continuously differentiable path $G : [0, 1] \rightarrow \mathfrak{E}_d$ satisfies*

$$(2.8) \quad d_N(\mathcal{M}_{G(0)}, \mathcal{M}_{G(1)}) \leq 4 \int_0^1 g_{N,G(t)}(G'(t)) dt.$$

Proof. We begin with an interior effect G , so that G and $I_d - G$ are invertible. Set $S = \sqrt{G(I_d - G)}$. For a Hermitian matrix K put $H_0 = SK + KS$. At the two factors $A_1 = \sqrt{G}$, $A_0 = \sqrt{I_d - G}$ prescribe derivatives

$$(2.9) \quad B_1 = \sqrt{I_d - G} K, \quad B_0 = -\sqrt{G} K.$$

They obey

$$(2.10) \quad \begin{aligned} A_1^* B_1 + B_1^* A_1 &= H_0, & A_0^* B_0 + B_0^* A_0 &= -H_0, \\ \sum_{y=0}^1 A_y^* B_y &= 0, & \sum_{y=0}^1 B_y^* B_y &= K^2. \end{aligned}$$

These tangent equations can be realized by genuine dilations. Let $G_1(s) = G + sH_0$, $G_0(s) = I_d - G - sH_0$ for s in a sufficiently small open interval, and put $R_y(s) = \sqrt{G_y(s)}$. The first line of (2.10) implies that

$$T_y = (B_y - R'_y(0))R_y(0)^{-1}$$

is anti-Hermitian. Indeed, multiplying $T_y + T_y^*$ on the left and right by $R_y(0)$ gives, with $R = R_y(0)$,

$$R(T_y + T_y^*)R = RB_y + B_y^*R - (RR'_y(0) + R'_y(0)R) = 0,$$

by differentiating $R_y(s)^2 = G_y(s)$. Thus $\tilde{A}_y(s) = \exp(sT_y)R_y(s)$ has derivative B_y at zero and satisfies $\tilde{A}_y(s)^* \tilde{A}_y(s) = G_y(s)$ exactly. A measurement isometry is

$$W(s)\psi = \sum_{y=0}^1 |y\rangle_C |y\rangle_L \tilde{A}_y(s)\psi.$$

Here C is the observed outcome, and L and the factor space of $\tilde{A}_y$ are environments. Tracing the environments implements $\mathcal{M}_{G(s)}$, also on inputs entangled with a reference. At zero, (2.10) gives

$$(2.11) \quad W^* \dot{W} = 0, \quad \dot{W}^* W = K^2.$$

The horizontal Kraus condition has the antecedents [22, Theorems 4–5]. Its adaptive channel-extension form is $F_N \leq 4N\|\alpha\|$ when $\beta = 0$; see [23, equation (9)] and [24, equations (5)–(10)]. In the present lift $\alpha = K^2$ and $\beta = 0$. We give the cancellation explicitly for this binary measurement.

Fix a common tester with N slots. Purify its initial state and all intervening operations, and retain the measurement environments only for this comparison; neither the outcome copy nor the factor environment is accessible to the tester. The derivative of its final purification in the direction H_0 is the sum of N terms, each having one differentiated occurrence of W . The terms are mutually orthogonal. For two different differentiated slots, cancel the common isometries after the later slot; the remaining overlap contains $W^* \dot{W} \otimes I = 0$. Each term has norm at most $\|\dot{W}\|_{\text{op}} = \|K\|_{\text{op}}$, independently of the reference and memory dimensions. Hence the derivative of the purification has norm at most $\sqrt{N}\|K\|_{\text{op}}$. The pure-state trace-norm derivative bound, followed by contraction under the partial trace, bounds the observed output derivative by

$$(2.12) \quad 2\sqrt{N}\|K\|_{\text{op}}.$$

A publicly stopped tester is padded to N slots by calls on fixed dummy inputs, whose outcomes and environments are both discarded. The same purification argument then applies to its original output. Classical feedback is included by purifying the common controlled operations.

Now let H be an arbitrary Hermitian tangent at G , and set $\kappa = N^{-1/2}$. There is a unique Hermitian solution of

$$(2.13) \quad SK + KS + \kappa K = H.$$

Write $H = H_0 + R$, with $H_0 = SK + KS$ and $R = \kappa K$. Both are legitimate tangents at the interior effect G . The derivative of a fixed tester's output is linear in the inserted channel tangent. The part arising from H_0 is bounded by (2.12). The part arising from R is bounded by $2N\|R\|_{\text{op}}$. To verify the latter bound, the one-call tangent has two opposite reference blocks

$$B_\rho = \text{tr}_A[(R \otimes I)\rho], \quad -B_\rho.$$

Duality against Hermitian contractions gives $\|B_\rho\|_1 \leq \|R\|_{\text{op}}$ for every input state. Insert this tangent into each of the N slots in turn; all subsequent common channels contract the Hermitian trace norm. This proves the bound for its sum. Thus the actual tangent H has output derivative bounded by

$$(2.14) \quad 2\sqrt{N}\|K\|_{\text{op}} + 2N\|R\|_{\text{op}} = 4\sqrt{N}\|K\|_{\text{op}}.$$

The splitting is used only for the differential at an interior effect; it does not postulate a physical mixture realizing the entire path.

In an eigenbasis of G , with $v_i = g_i(1 - g_i)$, equation (2.13) reads

$$K_{ij} = \frac{H_{ij}}{\sqrt{v_i} + \sqrt{v_j} + \kappa}.$$

Since $(\sqrt{v_i} + \sqrt{v_j} + \kappa)^2 \geq v_i + v_j + \tau$,

$$(2.15) \quad \sqrt{N}\|K\|_{\text{op}} \leq \sqrt{N}\|K\|_{\text{HS}} \leq g_{N,G}(H).$$

The integration of local distinguishability has direct antecedents: [25, Section III, equation (26), PDF version 3] treats the parallel path bound, and [26, Section IV.A, equations (14)–(22)] applies adaptive metrological estimates. Here the regularized binary Sylvester split and its residual produce the explicit integrand. For an interior path, integrate the resulting output derivative bound for each fixed tester, and then take the supremum. This proves (2.8) without exchanging a supremum with a derivative.

For a path in the closed body, apply the result to $G_\epsilon(t) = (1 - 2\epsilon)G(t) + \epsilon I_d$, where $0 < \epsilon < 1/2$. At fixed N , the form $g_{N,G}(H)$ is continuous on the whole body and bounded above by $N\|H\|_{\text{HS}}$. The integrals therefore converge as $\epsilon \downarrow 0$. The hybrid inequality $d_N(\mathcal{M}_E, \mathcal{M}_F) \leq 2N\|E - F\|_{\text{op}}$ gives continuity of the distances at both endpoints. Passing to the limit proves the asserted closed-body statement. $\square$

2.2. The midpoint comparison.

Proof of Theorem 2.1. For the upper bound use $G(t) = (1 - t)E + tF$. The map $G \mapsto V_G = G - G^2$ is operator concave. For $0 < t \leq 1/2$, write $G(t) = (1 - 2t)E + 2tM$ and discard the positive multiple of V_E in the concavity inequality. Applying the same argument from the other endpoint gives

$$(2.16) \quad V_{G(t)} \succeq a(t)V_M, \quad a(t) = 2 \min\{t, 1 - t\}.$$

For a positive matrix D and a Hermitian matrix X ,

$$\langle X, (L_D + R_D)X \rangle_{\text{HS}} = 2 \text{tr}(DX^2) \geq 0.$$

Thus left-plus-right multiplication preserves the stated order on the real Hilbert space of Hermitian matrices. Since $a(t) \leq 1$, $\mathcal{A}_{G(t)} \succeq a(t)\mathcal{A}_M$. Inverting positive operators reverses their order, whence

$$g_{N,G(t)}(H) \leq a(t)^{-1/2}Q_N(E, F).$$

The scalar factor has integral 2 on $[0, 1]$. Lemma 2.2 yields $d_N \leq 8Q_N$; the trace-norm bound $d_N \leq 2$ gives the stated upper estimate.

For the lower bound choose any orthonormal eigenbasis e_i of M and put

$$(2.17) \quad q_{ij} = |H_{ij}| \sqrt{\frac{N}{v_i + v_j + \tau}}, \quad Q_N^2 = \sum_{i,j} q_{ij}^2.$$

Repeated input e_i gives Bernoulli probabilities $s = m_i - H_{ii}/2$, $t = m_i + H_{ii}/2$. Their variances satisfy

$$\max\{s(1 - s), t(1 - t)\} \leq s(1 - s) + t(1 - t) \leq 2m_i(1 - m_i).$$

Lemma 7.2 therefore gives

$$(2.18) \quad d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F) \geq \frac{1}{64} \min\{1, q_{ii}\}.$$

For $i \neq j$, restrict each input to $\mathcal{S} = \text{span}\{e_i, e_j\}$. The resulting channels have the compressed effects $E_{\mathcal{S}}, F_{\mathcal{S}}$, including on inputs entangled with a retained reference. Their midpoint is $M_{\mathcal{S}} = \text{diag}(m_i, m_j)$. Write their spectral data as (p, q, u) and (p', q', v) in the notation of Section 7, with gaps r, r' and angle h when both gaps are positive. Put

$$V_E^{\mathcal{S}} = p(1 - p) + q(1 - q), \quad V_F^{\mathcal{S}} = p'(1 - p') + q'(1 - q').$$

For the compressed matrices the exact identity

$$V_{M_{\mathcal{S}}} = \frac{1}{2}V_{E_{\mathcal{S}}} + \frac{1}{2}V_{F_{\mathcal{S}}} + \frac{1}{4}(F_{\mathcal{S}} - E_{\mathcal{S}})^2$$

shows, on taking traces, that

$$(2.19) \quad v_i + v_j + \tau \geq \frac{1}{2}(V_E^S + \tau), \quad v_i + v_j + \tau \geq \frac{1}{2}(V_F^S + \tau).$$

The scalar part of $F_S - E_S$ has zero off-diagonal entry. Its traceless part has operator norm $|ru - r'v|/2$. Decomposing this vector along the smaller radius gives

$$(2.20) \quad \begin{aligned} |H_{ij}| &\leq \frac{1}{2}|r - r'| + \min\{r, r'\} \sin(h/2) \\ &\leq \frac{1}{2}(|p - p'| + |q - q'| + \min\{r, r'\}h). \end{aligned}$$

When one radius is zero, the angular term is zero and the same inequality holds without choosing its direction. Each endpoint variance is bounded by the appropriate V_E^S or V_F^S . Equations (2.19)–(2.20) therefore imply

$$(2.21) \quad q_{ij} \leq \frac{1}{\sqrt{2}} (T_N(p, p') + T_N(q, q') + \min\{K_N(p, q), K_N(p', q')\}h).$$

The nonadaptive assertion of Theorem 7.1 applied to this compressed pair yields

$$(2.22) \quad d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F) \geq \frac{1}{8192} \min\{1, q_{ij}\}.$$

Every such witness is a legitimate experiment for the original effects by a fixed isometric embedding of $\mathcal{S}$. Since $\max_{i,j} q_{ij} \geq Q_N/d$, taking the strongest of (2.18) and (2.22) proves the lower bound. For $d = 1$ only the diagonal argument is needed.

Finally, a one-use reference-assisted difference has output blocks B_ρ and $-B_\rho$, with $B_\rho = \text{tr}_A[(E - F) \otimes I]\rho$. Trace-norm duality against Hermitian contractions gives $\|B_\rho\|_1 \leq \|E - F\|_{\text{op}}$. An eigenstate of $E - F$ at an extremal eigenvalue attains equality. This proves (2.7). $\square$

Corollary 2.3 (Uniform comparison with nonadaptive experiments). *For all $E, F \in \mathfrak{E}_d$ and all $N \geq 1$,*

$$(2.23) \quad d_N(\mathcal{M}_E, \mathcal{M}_F) \leq 65536d d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F).$$

Here nonadaptive experiments may entangle all inputs and retain a reference; they use no feedback between calls. The dimensional constant comes from the maximum of d^2 weighted entries and is not optimized.

Proof. If $Q_N \leq 1$, divide the upper estimate $8Q_N$ by the lower estimate $Q_N/(8192d)$. If $Q_N \geq 1$, the upper estimate is 2 and the lower estimate is $1/(8192d)$, which gives a smaller constant. The identical pair needs no division. $\square$

2.3. All spectral ranks and local forms.

Lemma 2.4 (Spectral noncancellation in every rank). *Let $\lambda_1(E) \geq \dots \geq \lambda_d(E)$ and $\lambda_1(F) \geq \dots \geq \lambda_d(F)$ be the ordered spectra. For every $1 \leq k \leq d$,*

$$(2.24) \quad B_N(\lambda_k(E), \lambda_k(F)) \leq d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F).$$

For effects diagonal in the same ordered orthonormal basis,

$$(2.25) \quad d_N(\mathcal{M}_E, \mathcal{M}_F) \leq \sum_{k=1}^d B_N(\lambda_k(E), \lambda_k(F)).$$

Proof. Suppose $\lambda_k(E) \geq \lambda_k(F)$. The span of the first k eigenvectors of E intersects the orthogonal complement of the first $k - 1$ eigenvectors of F in a nonzero subspace. A unit vector in that intersection has success probabilities $s \geq \lambda_k(E) \geq \lambda_k(F) \geq t$ under the two hypotheses. The monotonicity in Lemma 7.2 implies $B_N(s, t) \geq B_N(\lambda_k(E), \lambda_k(F))$. Repeated use of this vector proves the first assertion; reverse the hypotheses for the opposite ordering. Multiplicities cause no change in the dimension argument.

For the upper bound, first measure the common eigenbasis and then produce a bit whose success probability is its corresponding eigenvalue. Supply N independent bits in each of the d rows before the experiment. Every common adaptive tester is a common channel of these row arrays, including its reference blocks and all publicly stopped histories. The tensor-product triangle inequality gives (2.25); unused row bits are discarded. $\square$

The following local comparison will permit volume estimates at every boundary stratum. It concerns matrices themselves and does not require a coordinate chart for eigenvectors.

Lemma 2.5 (Uniform local comparison of the variance forms). *Set $W_G = G(I_d - G) + \tau I_d/2$. For a Hermitian H , put $q = g_{N,G}(H)$. Then*

$$(2.26) \quad \|W_G^{-1/2} H W_G^{-1/2}\|_{\text{HS}} \leq 2q.$$

For either sign for which $G \pm H/2 \in \mathfrak{E}_d$, one has

$$(2.27) \quad \|W_G^{-1/2} (W_{G \pm H/2} - W_G) W_G^{-1/2}\|_{\text{op}} \leq q + \frac{3}{4}q^2.$$

In particular, for $q \leq 1/4$,

$$(2.28) \quad \frac{1}{2}W_G \preceq W_{G \pm H/2} \preceq 2W_G.$$

Thus, if either $Q_N(E, F) \leq 1/4$ or $g_{N,E}(F - E) \leq 1/4$, the midpoint and the fixed-endpoint quadratic forms are comparable within factors 2.

Proof. In an eigenbasis of W_G , with eigenvalues $w_i \geq \tau/2$,

$$\frac{w_i + w_j}{N w_i w_j} \leq 4.$$

Multiplying this inequality by $N|H_{ij}|^2/(w_i + w_j)$ and summing proves (2.26). Put $Z = W_G^{-1/2} H W_G^{-1/2}$. The exact expansion

$$W_{G \pm H/2} - W_G = \pm \frac{1}{2}(H - GH - HG) - \frac{1}{4}H^2$$

has normalized linear part $\pm((I_d - 2G)Z + Z(I_d - 2G))/4$, since G commutes with W_G . Its operator norm is at most $\|Z\|_{\text{op}}/2 \leq q$. The normalized quadratic part is $-ZW_G Z/4$. Since $\tau \leq 1$, one has $\|W_G\|_{\text{op}} \leq 3/4$, and its norm is at most $3q^2/4$. This proves (2.27) and hence (2.28). The same order holds for left plus right multiplication, and inversion gives the quadratic-form comparison. Apply it first at $G = M$ or at $G = E$, respectively, to obtain the final assertion. $\square$

3. DESCRIPTION ENTROPY IN ARBITRARY MATRIX DIMENSION

The intrinsic comparison of Theorem 2.1 permits the full effect body to be treated without choosing eigenvector coordinates near repeated spectra. Throughout this section the input dimension $d \geq 1$ is fixed. Constants may depend on d , but not on the integer horizon $N \geq 1$ or on the accuracy parameter.

Write $\mathcal{C}_N(\mathfrak{E}_d, \delta)$ for the covering number under d_N , with centres in the same legal memoryless ordered binary interface, and write $\mathcal{C}_N^{\text{rat}}$ when the real and imaginary parts of every matrix entry of each centre are rational.

Theorem 3.1 (Entropy of the full effect body). *For every integer $d \geq 1$ there are constants $c_d, C_d, \delta_d > 0$ such that, for all integers $N \geq 1$ and all $0 < \delta \leq \delta_d$,*

$$(3.1) \quad \begin{aligned} c_d N^{d^2/2} [\log(N+2)]^{\lfloor d/2 \rfloor} \delta^{-d^2} &\leq \mathcal{C}_N(\mathfrak{E}_d, \delta) \\ &\leq \mathcal{C}_N^{\text{rat}}(\mathfrak{E}_d, \delta) \\ &\leq C_d N^{d^2/2} [\log(N+2)]^{\lfloor d/2 \rfloor} \delta^{-d^2}. \end{aligned}$$

Consequently the optimum fixed-length reusable description has length

$$(3.2) \quad \frac{d^2}{2} \log_2 N + \lfloor d/2 \rfloor \log_2 \log(N+2) + d^2 \log_2(1/\delta) + O_d(1).$$

The estimates include all support and spectral multiplicities. The rational-centre assertion here follows from a covering argument. Theorem 6.1 gives a deterministic finite exact encoder and decoder attaining the same order in every dimension. Theorem 8.2 retains its separate direct construction in dimension two. Computational cost is accounted for separately from the number of transmitted bits.

For $d = 1$, the result is the Bernoulli covering law $N^{1/2}\delta^{-1}$. For $d = 2$, it recovers Theorem 8.1. The next two cases have orders $N^{9/2} \log(N+2)\delta^{-9}$ and $N^8[\log(N+2)]^2\delta^{-16}$. The logarithmic power counts independent nested approaches of paired eigenvalues to the two support endpoints. We first establish a uniform measure estimate for actual operational balls, and then evaluate the total measure.

3.1. Operational balls and a regularized volume form. Set $n = d^2$, $\tau = N^{-1}$ and

$$W_E = E(I - E) + \frac{\tau}{2}I.$$

As in Section 2, let

$$(3.3) \quad g_{N,E}(H)^2 = N \langle H, (L_{W_E} + R_{W_E})^{-1} H \rangle_{\text{HS}}$$

on the real Hilbert space of Hermitian matrices. Let $G_{N,E}$ denote this positive quadratic form in Hilbert–Schmidt orthonormal coordinates, and define

$$(3.4) \quad \rho_N(E) = \sqrt{\det G_{N,E}}, \quad d\mu_N(E) = \rho_N(E) dE,$$

The real orthonormal basis consists of e_{ii} and, for $i < j$, $(e_{ij} + e_{ji})/\sqrt{2}$ and $i(e_{ij} - e_{ji})/\sqrt{2}$; dE is Lebesgue measure in these coordinates. The cutoff makes ρ_N positive and continuous on the whole closed body for each N .

Lemma 3.2 (Uniform volume of operational balls). *For fixed d there are $a_d, A_d, r_d > 0$ such that*

$$(3.5) \quad a_d r^{d^2} \leq \mu_N(B_{d_N}(E, r)) \leq A_d r^{d^2}$$

for every $N \geq 1$, every $E \in \mathfrak{E}_d$, and every $0 < r \leq r_d$. Balls are taken inside $\mathfrak{E}_d$.

Proof. We make explicit the two local consequences of Lemma 2.5 that will be used. If H is Hermitian and $q = g_{N,E}(H)$, diagonalizing W_E gives

$$(3.6) \quad \|W_E^{-1/2} H W_E^{-1/2}\|_{\text{HS}}^2 \leq 4q^2,$$

since its eigenvalues w_i are at least $\tau/2$ and

$$\frac{w_i + w_j}{N w_i w_j} = \tau(w_i^{-1} + w_j^{-1}) \leq 4.$$

For any legal $E + tH$, direct expansion yields

$$W_{E+tH} - W_E = \frac{t}{2}((I - 2E)H + H(I - 2E)) - t^2 H^2.$$

The matrices E and W_E commute, $\|I - 2E\|_{\text{op}} \leq 1$, and $\|W_E\|_{\text{op}} \leq 3/4$. Thus the norm of the normalized difference is at most $2|t|q + 3t^2q^2$. For $|t| \leq 1$ and q below an absolute constant, it follows that

$$(3.7) \quad \frac{1}{2}W_E \preceq W_{E+tH} \preceq 2W_E.$$

Loewner comparison passes to $L_W + R_W$ and, by inversion, to the quadratic forms. It also gives

$$(3.8) \quad 2^{-n/2} \rho_N(E) \leq \rho_N(E + tH) \leq 2^{n/2} \rho_N(E).$$

Apply this comparison from an endpoint to the midpoint and from the midpoint to either endpoint. Theorem 2.1 then provides constants $A, B, r_0 > 0$, depending only on d , such that

$$(3.9) \quad d_N(E, F) \leq r \implies g_{N,E}(F - E) \leq Ar,$$

$$(3.10) \quad g_{N,E}(F - E) \leq Br \implies d_N(E, F) \leq r,$$

for legal E, F and $0 < r \leq r_0$. Decrease r_0 so that Ar_0 lies in the small-form range of (3.7).

Let ω_n be the Euclidean volume of the unit ball in $\mathbb{R}^n$. The ellipsoid $g_{N,E}(H) \leq u$ has Lebesgue volume $\omega_n u^n / \rho_N(E)$. Equations (3.9) and (3.8) therefore imply

$$\mu_N(B_{d_N}(E, r)) \leq 2^{n/2} \omega_n (Ar)^n.$$

For the lower bound we construct a full legal ellipsoid, including when E lies on a support boundary. Choose a small constant $a > 0$ depending only on d , set $s = ar\tau$, and move towards the scalar interior point:

$$(3.11) \quad E_s = (1 - 2s)E + sI.$$

Take r_0 sufficiently small that $s \leq 1/4$. Then $sI \preceq E_s \preceq (1 - s)I$. Since the displacement commutes with E , its form is bounded by

$$g_{N,E}(E_s - E)^2 = N \sum_i \frac{s^2(1 - 2\lambda_i)^2}{2\lambda_i(1 - \lambda_i) + \tau} \leq d \frac{s^2}{\tau^2} = da^2 r^2.$$

Choose $a\sqrt{d} \leq B/4$. By (3.10), $d_N(E, E_s) \leq r/4$.

Let $X = X^*$ satisfy $g_{N,E_s}(X) \leq br$, with $b > 0$ to be chosen. Write ℓ_i for the eigenvalues of E_s and $w_i = \ell_i(1 - \ell_i) + \tau/2$. Since $\ell_i \geq s$ and $w_i \leq \ell_i + \tau/2$, we have

$$\begin{aligned} \|E_s^{-1/2} X E_s^{-1/2}\|_{\text{HS}}^2 &\leq g_{N,E_s}(X)^2 \max_{i,j} \frac{w_i + w_j}{N \ell_i \ell_j} \\ &\leq b^2 r^2 \left(\frac{2\tau}{s} + \frac{\tau^2}{s^2} \right) = b^2 \left(\frac{2r}{a} + \frac{1}{a^2} \right). \end{aligned}$$

The same bound holds with $I - E_s$ in place of E_s , because $1 - \ell_i \geq s$ and $w_i \leq 1 - \ell_i + \tau/2$. Taking $b \leq a/4$ and $ar_0 \leq 1$ makes both operator norms at most $1/2$. Consequently every point of the entire ellipsoid

$$\mathcal{A}_s = \{E_s + X : g_{N,E_s}(X) \leq br\}$$

is a legal effect. Choose also $b \leq B/4$. Equation (3.10) and the triangle inequality give $\mathcal{A}_s \subset B_{d_N}(E, r)$. The density comparison about E_s and the exact ellipsoid volume now imply

$$\mu_N(B_{d_N}(E, r)) \geq 2^{-n/2} \omega_n (br)^n.$$

All choices depend only on d . This proves the assertion on every boundary stratum as well as in the interior. $\square$

3.2. The total regularized volume.

Proposition 3.3 (Nested endpoint accumulation). *For fixed $d \geq 1$, uniformly over $N \geq 1$,*

$$(3.12) \quad \mu_N(\mathfrak{E}_d) \asymp_d N^{d^2/2} [\log(N + 2)]^{\lfloor d/2 \rfloor}.$$

Proof. In a spectral basis, the quadratic form has weight $N/(2w_i)$ on the i th diagonal direction and weight $N/(w_i + w_j)$ on each of the two Hilbert–Schmidt orthonormal real directions belonging to $i < j$. Hence

$$(3.13) \quad \rho_N(E) = 2^{-d/2} N^{d^2/2} \prod_i w_i^{-1/2} \prod_{i < j} (w_i + w_j)^{-1}.$$

We use the classical spectral change of variables [28]. For a simple spectrum, differentiation of $E = U \operatorname{diag}(\lambda) U^*$ gives

$$U^*(dE)U = d \operatorname{diag}(\lambda) + [U^* dU, \operatorname{diag}(\lambda)].$$

Each complex off-diagonal pair therefore contributes $(\lambda_i - \lambda_j)^2$ to the real Jacobian. Integration over the compact unitary orbit and over permutations supplies a positive constant depending only on d . The repeated-spectrum set is Lebesgue null. This spectral change of variables, together with (3.13), yields

$$(3.14) \quad \mu_N(\mathfrak{E}_d) \asymp_d N^{d^2/2} I_d(\tau),$$

where

$$(3.15) \quad I_d(\tau) = \int_{[0,1]^d} \prod_i (v_i + \tau)^{-1/2} \prod_{i < j} \frac{(\lambda_i - \lambda_j)^2}{v_i + v_j + \tau} d\lambda, \quad v_i = \lambda_i(1 - \lambda_i).$$

Replacing $v_i + \tau/2$ by $v_i + \tau$ in the single-coordinate factors changes the integral by a factor between fixed positive constants depending only on d .

We prove

$$(3.16) \quad I_d(\tau) \asymp_d [\log(2 + \tau^{-1})]^{[d/2]}, \quad 0 < \tau \leq 1.$$

For the upper bound, partition each coordinate into $[0, 1/4]$, $[1/4, 3/4]$, and $[3/4, 1]$. Factors involving an eigenvalue in the middle interval are bounded above by constants, so these coordinates can be integrated out. For each of the remaining $l \leq d$ coordinates put $x_i = \lambda_i$ near zero and $x_i = 1 - \lambda_i$ near one. On a sector $0 \leq x_1 \leq \dots \leq x_l \leq 1/4$, let $\epsilon_i \in \{1, -1\}$ record the endpoint and put

$$S_j = \sum_{i=1}^j \epsilon_i, \quad S_0 = 0, \quad y_j = x_j + \tau.$$

The individual factor is at most $Cy_j^{-1/2}$. For $i < j$, a same-endpoint pair factor is at most Cy_j , whereas an opposite-endpoint factor is at most C/y_j . The sector integrand is consequently bounded by

$$(3.17) \quad C_d \prod_{j=1}^l y_j^{a_j-1}, \quad a_j = \frac{1}{2} + \epsilon_j S_{j-1} = \frac{S_j^2 - S_{j-1}^2}{2}.$$

Its cumulative exponents satisfy

$$(3.18) \quad A_j = \sum_{i=1}^j a_i = \frac{S_j^2}{2} \geq 0.$$

Set $T = 1/4 + \tau$ and $L = \log(T/\tau)$. Substitute $y_j = T \exp(-z_j)$, where $z_1 \geq \dots \geq z_l \geq 0$, and then $u_j = z_j - z_{j+1}$ for $j < l$, $u_l = z_l$. The sector integral is bounded by

$$(3.19) \quad C_d \int_{u_j \geq 0, \sum_j u_j \leq L} \exp\left(-\sum_j A_j u_j\right) du.$$

The omitted factor T^{A_l} is bounded above uniformly in $0 < \tau \leq 1$. Each $A_j > 0$ is at least $1/2$, so its variable has an integrable exponential tail. Each vanishing A_j contributes at most one

factor $1 + L$. A vanishing prefix sum requires an even j , and there are at most $\lfloor l/2 \rfloor$ such indices. Summing the finitely many sectors proves the upper estimate in (3.16). The case $l = 0$ contributes only a bounded constant.

For the lower bound let $m = \lfloor d/2 \rfloor$. Choose scales

$$(3.20) \quad \tau \leq t_1, \quad 8t_j \leq t_{j+1} \ (j < m), \quad t_m \leq 1/32,$$

and restrict $2m$ eigenvalues in pairs by

$$(3.21) \quad \lambda_j^- = a_j \in [t_j, 2t_j], \quad \lambda_j^+ = 1 - b_j, \quad b_j \in [t_j, 2t_j], \quad 1 \leq j \leq m.$$

If d is odd, restrict the remaining eigenvalue to $[3/8, 5/8]$; all its factors are bounded above and below by positive constants. For one pair the two individual spectral factors and its opposite-endpoint factor have product comparable to t_j^{-2} . For two pairs at scales $t_i \leq t_j/8$, the two same-endpoint factors are comparable to t_j and the two opposite-endpoint factors to t_j^{-1} . Their product is therefore bounded above and below by fixed constants. Thus the integrand on the box is comparable to $\prod_j t_j^{-2}$ and its volume to $\prod_j t_j^2$. Each box contributes at least $c_d > 0$.

Take the scales from $\tau, 8\tau, 8^2\tau, \dots$ up to $1/32$. The boxes for distinct increasing m -tuples are disjoint. Their number is a binomial coefficient bounded below by $c_d [\log(\tau^{-1})]^m$ once $\log(\tau^{-1})$ is sufficiently large depending on d . In the remaining bounded range, a fixed box of distinct eigenvalues strictly inside $(0, 1)$ has a positive uniform integral and supplies the lower estimate. For $m = 0$ this interior argument suffices on the full range. This proves (3.16) and, by (3.14), the proposition. $\square$

The balanced-prefix identity (3.18) accounts for all nested endpoint scales. A simultaneous approach of all eigenvalues to the endpoints detects only one possible critical scaling; the separated pairs in (3.21) show why the logarithmic multiplicity can be larger.

3.3. Covering, rational centres, and description length.

Proof of Theorem 3.1. Choose δ_d smaller than the radius in Lemma 3.2. A cover by L balls of radius δ satisfies

$$\mu_N(\mathfrak{E}_d) \leq LA_d \delta^{d^2},$$

which gives the required lower bound. Conversely, take a maximal δ -separated subset of $\mathfrak{E}_d$. Its balls of radius $\delta/3$ are pairwise disjoint, and the lower bound in (3.5) bounds its cardinality by

$$C_d \mu_N(\mathfrak{E}_d) \delta^{-d^2}.$$

It is a radius- δ cover by maximality. Such finite maximal sets exist: the effect body is compact, d_N is continuous by the hybrid bound, and $d_1(E, F) = 2\|E - F\|_{\text{op}}$ separates distinct effects. Proposition 3.3 now gives both estimates for legal centres.

Rational legal effects are dense. Indeed, first move towards $I/2$ by (3.11), and then approximate all independent real matrix coordinates by rational numbers while remaining strictly between 0 and I . For fixed N , continuity in the operational distance follows again from the hybrid bound. Replace every centre of a radius- $\delta/2$ cover by a rational legal effect at distance less than $\delta/2$. The resulting radius- δ cover has the same cardinality. This proves the rational-centre upper bound.

A fixed-length payload indexes the centres of a cover, and conversely each legal reusable program with a fixed payload supplies a centre. Taking base-two logarithms of (3.1) and rounding the index length gives (3.2). $\square$

4. LEARNING THE COMPLETE BINARY QUBIT EFFECT BODY

We now pass from supplied descriptions to an unknown device. Throughout this section the input dimension is two. Write

$$E = \frac{1}{2}((1+b)I + ru \cdot \sigma), \quad 0 \leq r \leq 1, \quad |b| + r \leq 1,$$

with the direction absent when $r = 0$. The bias, contrast, and direction are all unknown. The outputs are ordered and classical, and each call consumes its input without a residual quantum output. A learning procedure may prepare entangled input blocks, retain the unused input qubits, and use classical feedback between blocks. It has no access to a measurement environment. Its choices depend only on the public parameters and the observed record.

The integer N below is the horizon of a future comparison, measured by d_N . The number M of training calls is a separate resource. The output is a classical description of a legal effect. For a unit vector w set $P_w = (I + w \cdot \sigma)/2$ and write $\mathcal{M}_{p,q,w}$ for the measurement associated with $pP_w + q(I - P_w)$, where $p = (1 + b + r)/2$ and $q = (1 + b - r)/2$. We use the spectral quantities

$$V = p(1-p) + q(1-q), \quad K_N(p, q) = r\sqrt{N/(V + N^{-1})}.$$

Theorem 4.1 (Common learning at the finite-use scale). *There are absolute constants $c, C, \delta_0 > 0$ such that, for every $N \geq 1$, $0 < \delta \leq \delta_0$, and $0 < \eta \leq 1/8$, one common learning procedure returns an effect $\hat{E} \in \mathfrak{E}_2$ with*

$$(4.1) \quad \sup_{E \in \mathfrak{E}_2} \mathbb{P}_E\{d_N(\mathcal{M}_E, \mathcal{M}_{\hat{E}}) > \delta\} \leq \eta.$$

On every record it uses at most

$$(4.2) \quad M \leq CN\delta^{-2} \log(1/\eta)$$

calls, and every entangled input block has length at most N . Conversely, every common procedure satisfying (4.1), with at most M calls on every record, obeys

$$(4.3) \quad M \geq cN\delta^{-2} \log(1/\eta).$$

The converse permits arbitrary reference-assisted adaptive training and bounded public stopping. The constants are uniform over the closed effect body, including scalar, deterministic, and projective effects.

The budget in (4.2) is a deterministic upper bound for each completed record, including unsuccessful records. It counts calls during training; N specifies only the subsequent loss. The construction uses independent input blocks, retaining at most $N - 1$ unused input qubits after the next qubit has been supplied, and no quantum system between blocks. Its preparation model permits ideal trusted finite-state preparation and fresh fair random signs. Preparation cost, classical running time, and classical workspace are separate resources. The reusable payload is charged separately in Corollary 4.6. The proof below specifies a written statistical algorithm; the accompanying finite identity checks do not constitute a software implementation of its risk guarantee.

The proof constructs one decision tree for the entire parameter body. Its experiments are not selected from a known pair of hypotheses. The angular part uses multiscale phase estimation, a method with antecedents in [31, 32]. We give the phase and confidence bounds explicitly. A data-dependent amplitude threshold selects the block length without prior calibration of the spectrum.

4.1. Bernoulli errors and estimated axes. For $s, t \in [0, 1]$ put

$$(4.4) \quad \mathfrak{h}(s, t)^2 = (\sqrt{s} - \sqrt{t})^2 + (\sqrt{1-s} - \sqrt{1-t})^2.$$

Write $B_N(s, t)$ for the unhalved distance between the two N -fold Bernoulli product laws. We use the following elementary estimates at both support endpoints.

Lemma 4.2 (Endpoint estimation). *For every $s, t \in [0, 1]$,*

$$(4.5) \quad B_N(s, t) \leq 2\sqrt{N} \mathfrak{h}(s, t).$$

For each fixed $A > 0$ there is C_A such that, if $0 < \lambda \leq 1$ and

$$(4.6) \quad |s - t| \leq A(\sqrt{t(1-t)\lambda} + \lambda),$$

then $\mathfrak{h}(s, t) \leq C_A \sqrt{\lambda}$. If two estimates of an ordered pair (p, q) each have $\mathfrak{h}$ error at most ε , sorting the estimates preserves this bound up to an absolute constant.

Proof. The root fidelity is $1 - \mathfrak{h}^2/2$. Tensorization and the trace-fidelity inequality give

$$B_N(s, t) \leq 2\sqrt{1 - (1 - \mathfrak{h}(s, t)^2/2)^{2N}} \leq 2\sqrt{N} \mathfrak{h}(s, t).$$

For the first square root in (4.4), if $t \geq \lambda$ divide the difference by $\sqrt{t}$ in (4.6). If $t < \lambda$, both t and s are at most $C_A \lambda$. Apply the same argument to their complements. For sorting, set $\phi(t) = \arcsin \sqrt{t}$. Then

$$\mathfrak{h}(s, t) = 2 \sin(|\phi(s) - \phi(t)|/2).$$

The two distances are uniformly comparable on $[0, \pi/2]$. Sorting is nonexpansive for the sup norm against an already ordered pair of real numbers. Apply this fact in the ϕ coordinate. $\square$

If $\hat{t}$ is the empirical frequency in n Bernoulli trials, Bernstein's inequality gives, with failure at most $2e^{-L}$,

$$(4.7) \quad |\hat{t} - t| \leq \sqrt{2t(1-t)L/n} + \frac{2L}{3n}.$$

Thus Lemma 4.2 applies with $\lambda = L/n$, including at $t = 0, 1$.

For a unit estimate w of u , write

$$(4.8) \quad h = \angle(u, w), \quad \gamma = \frac{r}{2}(1 - u \cdot w).$$

The input states on the axes $+w$ and $-w$ have respective success probabilities

$$(4.9) \quad s_+ = p - \gamma, \quad s_- = q + \gamma, \quad p = \frac{1}{2}(1 + b + r), \quad q = \frac{1}{2}(1 + b - r).$$

If $\hat{x}$ estimates $x = ru$ with Euclidean error at most e , normalizing $\hat{x}$ gives

$$(4.10) \quad rh \leq \pi \min(r, e), \quad \gamma \leq \min(r, e^2/r) \quad (r > 0).$$

Indeed $\|u - w\| \leq 2e/r$, $h \leq (\pi/2)\|u - w\|$, and $\gamma = r\|u - w\|^2/4$. If $\hat{x} = 0$, choose a fixed direction; then $e \geq r$ gives the same bounds. At $r = 0$ both quantities vanish.

Lemma 4.3 (Product learning away from unit contrast). *Suppose $r \leq 7/8$. There is a product-input procedure, not requiring b, r, u as input, which uses eight groups of n calls and returns a legal effect F such that, for $1 \leq L \leq n$,*

$$(4.11) \quad d_N(\mathcal{M}_E, \mathcal{M}_F) \leq C(\sqrt{NL/n} + NL/n)$$

outside an event of probability at most Ce^{-L} .

Proof. First use the positive and negative eigenstates of each of the three Pauli axes, with n calls per state. Encode an observed outcome as $Z \in \{-1, 1\}$. The two means on axis j are $b + x_j$ and $b - x_j$. Estimate x_j by half their empirical difference. Put

$$w_0 = 1 - b^2, \quad \lambda = L/n.$$

The two variances on axis j sum to $2(w_0 - x_j^2)$. Bernstein's inequality and a union bound therefore give

$$(4.12) \quad e = \|\hat{x} - x\| \leq 3(\sqrt{w_0\lambda} + \lambda).$$

More explicitly, the coordinate error is at most $\sqrt{2w_0\lambda} + 4\lambda/3$; the sign observation has range length two. Taking the Euclidean norm gives (4.12), since $\sqrt{6} < 3$ and $4\sqrt{3}/3 < 3$.

Normalize to a direction w . Feasibility and $r \leq 7/8$ imply

$$(4.13) \quad w_0 \geq r(2 - r), \quad V = \frac{1}{2}(w_0 - r^2) \geq w_0/9.$$

Use (4.10) and Theorem 7.4. The square-root term in (4.12) contributes at most $C\sqrt{N\lambda}$ to $K_N(p, q)h$, and its additive term contributes at most $CN\lambda$. Consequently,

$$(4.14) \quad d_N(\mathcal{M}_{p,q,u}, \mathcal{M}_{p,q,w}) \leq C(\sqrt{N\lambda} + N\lambda).$$

This calculation does not require a lower bound on r or w_0 .

It remains to control both endpoint probabilities in (4.9). For either $t = p$ or $t = q$ put $v_t = t(1 - t)$. Since

$$|p(1 - p) - q(1 - q)| = r|b| \leq r, \quad w_0 = 2V + r^2 \leq 4v_t + 3r,$$

we claim

$$(4.15) \quad \gamma \leq 9\sqrt{v_t\lambda} + 72\lambda.$$

Equation (4.12) gives $e^2 \leq 18(w_0\lambda + \lambda^2)$. If $r \leq \lambda$, use $\gamma \leq r$. If $r > \lambda$, use

$$\gamma \leq 72v_t\lambda/r + 72\lambda, \quad \gamma \leq r,$$

and $\min(r, 72v_t\lambda/r) \leq \sqrt{72v_t\lambda}$. This proves (4.15).

Make two fresh groups of n calls, along $+w$ and $-w$. Conditional on the previous record, these are Bernoulli trials with the parameters in (4.9). Their variances differ from v_t by at most γ . Equations (4.7) and (4.15) imply

$$|\hat{t}_{\text{raw}} - t| \leq C(\sqrt{v_t\lambda} + \lambda).$$

For the additional term use $\sqrt{\gamma\lambda} \leq C(\sqrt{v_t\lambda} + \lambda)$, which follows from $\sqrt{AB} \leq (A + B)/2$ with $A = \sqrt{v_t\lambda}$ and $B = \lambda$. Lemma 4.2 gives root-distance error $C\sqrt{\lambda}$ for both endpoints. Sort the two empirical parameters to obtain $\hat{p} \geq \hat{q}$ and return $F = \hat{p}P_w + \hat{q}(I - P_w)$. The aligned programme bound (7.9) and (4.5) give a spectral contribution at most $C\sqrt{N\lambda}$. Combine with (4.14). All concentration bounds for the fresh groups hold conditionally, so their failure probabilities may be added to those of the first six groups. $\square$

4.2. A common entangled experiment with unknown bias.

Lemma 4.4 (Two-plane phase observation). *Fix an orthonormal frame (e_1, e_2, w) known to the experimenter. For each $i = 1, 2$ and integer $m \geq 1$, there is an m -call experiment with an observation $Y \in \{-1, 1\}$ whose mean, for a chosen phase φ , is*

$$(4.16) \quad \mathbb{E}_E Y = \Re(e^{i\varphi} Z_i), \quad Z_i = [x \cdot w + ix \cdot e_i]^m, \quad x = ru.$$

Two phases estimate the complex number Z_i . For $0 < \zeta \leq 1$ and $0 < \epsilon \leq 1/2$, four settings, using the two planes, estimate both complex numbers with error at most ζ and failure at most ϵ , with at most $C\zeta^{-2} \log(C/\epsilon)$ independent blocks per setting.

Fix $a = 1/64$. If $r > 0$ and $h = \angle(u, w) \leq 2a/m$, then

$$(4.17) \quad .99r^m \leq |Z_i| \leq r^m, \quad \text{Arg } Z_i = m \arctan \frac{u \cdot e_i}{u \cdot w},$$

where the principal argument is used. If the amplitudes are bounded below by a fixed positive constant and ζ is sufficiently small, these observations produce a direction estimate w' with

$$(4.18) \quad \angle(u, w') \leq C\zeta/m.$$

Proof. Choose a computational basis along the axis perpendicular to the plane spanned by w, e_i , with phase convention such that the off-diagonal matrix element of $O = 2E - I = bI + ru \cdot \sigma$ is $r(u \cdot w + iu \cdot e_i)$. Prepare

$$|G_{\varphi, s}\rangle = \frac{|0\rangle^{\otimes m} + se^{i\varphi}|1\rangle^{\otimes m}}{\sqrt{2}},$$

where s is a fresh fair sign for this block. Conditional on the entire preceding record, the two signs have probability $1/2$ each; the random draw is independent of all earlier preparation randomness. Pass its qubits through the measurement and record $Y = s \prod_{j=1}^m Z_j$, where each output is encoded as $Z_j \in \{-1, 1\}$. Memorylessness makes the joint outcome law the product POVM on the entangled input; the qubits may be supplied sequentially while the unused ones are retained.

The expansion of $\langle G_{\varphi, s} | O^{\otimes m} | G_{\varphi, s} \rangle$ has diagonal term

$$\frac{1}{2}((b + ru \cdot f_i)^m + (b - ru \cdot f_i)^m),$$

where f_i is the chosen perpendicular axis. It is independent of s and disappears after multiplication by s and averaging. The off-diagonal terms give (4.16). At $x = 0$ the mean is zero and no direction is needed. In particular the experiment removes the unknown bias exactly. The phases $0, \pi/2$ give the real part and the negative imaginary part. More precisely, at each of the four settings use

$$(4.19) \quad B(\zeta, \epsilon) = \left\lceil 4\zeta^{-2} \log(8/\epsilon) \right\rceil$$

fresh independent blocks. Hoeffding's inequality bounds the error of each real sample mean by $\zeta/\sqrt{2}$ outside an event of probability $2\exp(-B\zeta^2/4)$. The union bound over the four means is at most ϵ , and each complex error is at most ζ . This bound holds conditional on every preceding record: the frame and the setting are then fixed, and the memoryless device acts on fresh blocks. Each estimate batch uses its prescribed number of blocks without stopping within the batch.

For the phase conclusions assume $r > 0$ and put $\theta_i = \arctan((u \cdot e_i)/(u \cdot w))$. Then $|m\theta_i| \leq 2a$ and

$$|Z_i| = r^m(1 - (u \cdot f_i)^2)^{m/2} \geq r^m \cos(h)^m \geq r^m(1 - 2a^2/m).$$

This proves (4.17), with no phase ambiguity. For the estimates set

$$\hat{\theta}_i = \text{Arg}(\hat{Z}_i)/m, \quad w' = \frac{w + \tan(\hat{\theta}_1)e_1 + \tan(\hat{\theta}_2)e_2}{\|w + \tan(\hat{\theta}_1)e_1 + \tan(\hat{\theta}_2)e_2\|}.$$

The exact formula with θ_i recovers u . For $|\hat{Z}_i - Z_i| \leq |Z_i|/2$,

$$|\text{Arg}(\hat{Z}_i) - \text{Arg}(Z_i)| \leq 2|\hat{Z}_i - Z_i|/|Z_i|.$$

On $|\theta_i| \leq 1/4$ the displayed map from two angles to a unit vector is Lipschitz; a bound of 16 for angular distance against the largest coordinate error follows from differentiation of $\tan$ and normalization. This proves (4.18). Our fixed cone lies strictly inside the principal-argument interval, and the argument uses no unwrapping threshold from a separate phase-estimation theorem. $\square$

We make the reconstruction rule total. Fix a coordinate basis of $\mathbb{R}^3$ and a tie-breaking order. Given a unit vector w , choose the first coordinate vector minimizing its absolute inner product with w , normalize its projection onto $w^\perp$ to obtain e_1 , and put $e_2 = w \times e_1$. The projection has norm at least $\sqrt{2/3}$, so the frame is defined at every w . For complex estimates z_1, z_2 use the phase formula in the preceding proof only when both pass the guard

$$(4.20) \quad \Re z_i > 0, \quad 8|\Im z_i| \leq \Re z_i \quad (i = 1, 2).$$

Write $\mathcal{R}_m(w; z_1, z_2) = (\text{ok}, w')$ for the resulting update, and set $\mathcal{R}_m = (\text{fail}, w)$ otherwise. A passed guard gives $|\text{Arg } z_i| \leq \arctan(1/8) < 1/4$; the vector to be normalized has inner product one with w and cannot vanish. Zero complex estimates, the branch cut, and an out-of-cone phase therefore have prescribed outcomes before an argument or tangent is evaluated.

Lemma 4.5 (Selecting a block length from the observations). *Suppose $r \geq 5/8$ and an initial known direction w satisfies $\angle(u, w) \leq a = 1/64$. A common adaptive procedure selects a length m_* and a direction w_* , for $N \geq 1$ and $0 < \eta \leq 1/8$, using at most $CN \log(C/\eta)$ calls on every record. With failure at most $\eta/16$,*

$$(4.21) \quad \frac{1}{8} \min\{N, (1-r)^{-1}\} \leq m_* \leq N,$$

$$(4.22) \quad r^{m_*} \geq 1/5, \quad \angle(u, w_*) \leq a/m_*.$$

The reciprocal at $r = 1$ is interpreted as infinity. New observations at m_*, w_* have both complex amplitudes at least $1/6$.

Proof. Fix $\kappa = a/1024$, and let H be the largest power of two not exceeding N . At a tested length m , estimate both complex numbers in Lemma 4.4 to error at most κ with $B(\kappa, \epsilon_m)$ blocks per setting, where

$$(4.23) \quad \epsilon_m = \frac{\eta m}{32N}.$$

The following procedure stops at the first rejection; it never tests a larger length after that rejection. Here w_0 is the input direction and $\perp$ denotes an empty register.

SELECTBLOCK(N, η, w_0)

1. Set $m \leftarrow 1$, $w \leftarrow w_0$, and $A \leftarrow \perp$.
2. Obtain z_1, z_2 from the four fresh settings at length m and frame w , with budgets (4.23).
3. If $\min_i |z_i| < 1/4$, return A if $A \neq \perp$, and return $(1, w_0)$ otherwise.
4. Set $(s, v) \leftarrow \mathcal{R}_m(w; z_1, z_2)$. If $s = \text{fail}$, return A if $A \neq \perp$, and return $(1, w_0)$ otherwise.
5. Set $A \leftarrow (m, v)$. If $m = H$, return A .
6. Set $w \leftarrow v$, $m \leftarrow 2m$, and return to step 2.

Equality in the amplitude threshold is accepted, and equality in the second phase guard is allowed. All registers, decisions, and fallbacks are functions of the observed record.

We prove the assertions on the event that the successive complex estimates obey their bounds. Before a length m is tried the angular error is at most $2a/m$: this holds initially, and is preserved by each accepted update. If the amplitude test passes, each true amplitude $\rho = |Z_i|$ is at least $1/4 - \kappa > 1/5$. Since $|\text{Arg } Z_i| \leq 2a$,

$$\Re z_i \geq \rho(1 - 2a^2) - \kappa > 8(2a\rho + \kappa) \geq 8|\Im z_i|.$$

The strict inequality follows directly from $a = 1/64$, $\kappa = a/1024$, and $\rho \geq 1/4 - \kappa$. Thus the guard cannot reject such a stage on this event. The argument bound and the Lipschitz constant in Lemma 4.4 give an updated angular error at most

$$\frac{32\kappa}{(1/4 - \kappa)m} \leq \frac{a}{4m}.$$

Furthermore $r^m \geq 1/4 - \kappa$, whence $m(1 - r) \leq 2$.

On rejection, the amplitude test must have failed, and (4.17) gives $r^m < (1/4 + \kappa)/.99 < 1/3$. Since $r \geq 5/8$,

$$-\log r \leq (1-r)/r \leq 2(1-r), \quad m(1-r) \geq (\log 3)/2.$$

The procedure returns the preceding accepted length $m/2$, which is therefore at least $(\log 3)/(4(1-r))$. Length one is always accepted on the good event, since its true amplitudes exceed $.99(5/8)$. If the cap is reached, the selected length is $H \geq N/2$. These facts prove (4.21) and (4.22). At the updated direction the amplitude is at least $r^{m*} \cos(a/m_*)^{m*} > 1/6$.

For completeness, let $\mathcal{F}_m$ contain the record immediately before a possible stage m , let $T_m \in \mathcal{F}_m$ be the event that this stage is reached, and let G_m be its complex-error event. On every reached record the concentration bound gives $\mathbb{P}_E(G_m^c | \mathcal{F}_m) \leq \epsilon_m$. Consequently

$$(4.24) \quad \mathbb{P}_E\left(\bigcup_m (T_m \cap G_m^c)\right) \leq \sum_m \mathbb{E}_E[\mathbf{1}_{T_m} \mathbb{P}_E(G_m^c | \mathcal{F}_m)] \leq \sum_{m=1,2,4,\dots,H} \epsilon_m \leq \eta/16.$$

The sample-mean bound holds for every preceding record, even one whose frame has left the cone; the cone is needed only to turn those sample means into directional information. Independence between stage-failure events is not assumed. By Lemma 4.4, the number of calls is bounded by an absolute constant times

$$(4.25) \quad \sum_{m=1,2,4,\dots,H} m \log \frac{CN}{\eta m} \leq C' N \log(C'/\eta).$$

To see this, write $m = H/2^k$ and use the convergent sums of 2^{-k} and $k2^{-k}$, together with $1 \leq N/H < 2$. The public cap and these budgets bound the cost on every record, including records on which an estimate fails. Rejection can only shorten the sum. $\square$

4.3. The learning bound and its converse. All statistical batches use fresh inputs and fresh randomization. Table 1 allocates their error probabilities. Each conditional bound is uniform over the preceding record. A row belonging to a branch not taken contributes no failure event.

TABLE 1. Confidence allocation for one execution of the learner.

Batch	Failure allowance	Conditioning and scope
Coarse Pauli vector	$\eta/8$	Unconditional; all six input groups jointly.
Product-branch Pauli vector	$\eta/8$	Given the coarse record; six fresh input groups jointly.
Dyadic stage m	$\eta m/(32N)$	Given the record before that stage; all reached stages together have allowance at most $\eta/16$.
Fine two-plane estimate	$\eta/8$	Given the complete search record, including the selected m_*, w_* .
Fresh endpoint groups	$\eta/16$ each	Given the final estimated axis; the two allowances sum to $\eta/8$, in either branch.

The conservative sum of all rows is $9\eta/16 < \eta$. The conditional union calculation (4.24) applies to every row, so this sum requires no independence between the events attached to different batches.

Proof of Theorem 4.1. We choose constants in the following order. Fix the local cone $a = 1/64$ and $\kappa = a/1024$ as above. Choose a sufficiently small absolute constant $c_* > 0$ in the final phase tolerance below. Then choose the absolute multipliers in the Bernoulli sample counts sufficiently

large. Finally reduce δ_0 , if needed, so that $\delta_0 \leq 1/4096$ and $c_*\delta_0 \leq a/100$. No optimized value of these constants is used. Set

$$(4.26) \quad L = \log(C_2/\eta), \quad n_0 = \lceil C_0 L \rceil, \quad n = \lceil C_1 N \delta^{-2} L \rceil,$$

where C_0, C_1, C_2 are absolute constants large enough for the concentration bounds and loss estimates below. The common procedure is the following.

- (1) Make the six coarse Pauli experiments, with n_0 calls per input, and form their half-difference vector $\hat{x}_0$.
- (2) If $\|\hat{x}_0\| < 3/4$, make six fresh Pauli groups of n calls per input and form $\hat{x}$. Set $w_f = \hat{x}/\|\hat{x}\|$, using the fixed third coordinate direction if $\hat{x} = 0$, and proceed to step 4.
- (3) Otherwise set $w_0 = \hat{x}_0/\|\hat{x}_0\|$ and run $\text{SELECTBLOCK}(N, \eta, w_0)$. At its returned (m_*, w_*) , obtain two fresh complex estimates with tolerance $\zeta_* = c_*\delta\sqrt{m_*/N}$ and failure allowance $\eta/8$. Apply $\mathcal{R}_{m_*}$ to them; take its updated direction as w_f on success and retain $w_f = w_*$ on failure.
- (4) Make n fresh calls along each of $+w_f$ and $-w_f$. If the empirical success frequencies are s_+, s_- , return

$$\hat{E} = \max(s_+, s_-)P_{w_f} + \min(s_+, s_-)(I - P_{w_f}).$$

The empirical frequencies already lie in $[0, 1]$. Ties in the coarse comparison use step 3; ties between endpoint frequencies give a scalar effect. Together with the frame, phase, and first-rejection conventions above, these rules define an output for every record, with no retry or additional calibration step.

We verify the risk. With failure at most $\eta/8$, the coarse half-difference estimate satisfies

$$\|\hat{x}_0 - x\| \leq \varepsilon_0, \quad \varepsilon_0 \leq a/100.$$

On this event the two decisions have the explicit overlap

$$\begin{aligned} \|\hat{x}_0\| < 3/4 &\implies r < 3/4 + 1/6400 < 7/8, \\ \|\hat{x}_0\| \geq 3/4 &\implies r \geq 3/4 - 1/6400 > 5/8. \end{aligned}$$

For the product branch, Lemma 4.3 gives error at most $\delta/2$ with the sample sizes in (4.26), after increasing C_1 . Its first six groups have joint allowance $\eta/8$ conditional on the coarse record. After their axis is fixed, each endpoint group has allowance $\eta/16$. These are the product-branch and endpoint rows of Table 1, and all samples here are fresh relative to the coarse test.

For the entangled branch, the normalized initial estimate has angular error at most $\pi\varepsilon_0/r \leq 8\pi a/500 < a$ by (4.10). Apply Lemma 4.5. At the selected length and updated direction, make fresh two-plane observations with complex error tolerance

$$(4.27) \quad \zeta_* = c_*\delta\sqrt{m_*/N},$$

and failure allowance $\eta/8$, conditional on the full selection record. On the preceding good event, the two true amplitudes are at least $1/6$ and their phases have absolute value at most a . On the current complex-error event, $\zeta_* \leq a/100$ and the inequalities $|\sin t| \leq |t|$ and $\cos t \geq 1 - t^2/2$ show that both phase guards pass. The resulting direction w_f satisfies

$$(4.28) \quad h_f = \angle(u, w_f) \leq Cc_*\delta/\sqrt{Nm_*}.$$

The calls used in this step are at most

$$(4.29) \quad Cm_*\zeta_*^{-2} \log(C/\eta) \leq C'N\delta^{-2} \log(C/\eta).$$

Put $\nu = 1 - r$. Since $|b| \leq \nu$ and $r \geq 5/8$,

$$r\nu \leq V = \frac{1}{2}(1 - r^2 - b^2) \leq \nu.$$

Equation (4.21) implies

$$(4.30) \quad m_*(V + N^{-1}) \geq c_0 > 0$$

for an absolute c_0 . Thus (4.28) and Theorem 7.4 give

$$(4.31) \quad d_N(\mathcal{M}_{p,q,u}, \mathcal{M}_{p,q,w_f}) \leq Cc_*\delta.$$

The endpoint misalignment is also bounded:

$$(4.32) \quad \gamma = \frac{r}{2}(1 - u \cdot w_f) \leq rh_f^2/4 \leq Cc_*^2\delta^2/N.$$

The final fresh groups have the public size n in (4.26). Conditional on the record fixing w_f , their success probabilities are exactly $p - \gamma$ and $q + \gamma$, and each group obeys Bernstein's bound with allowance $\eta/16$. Conditional Bernstein bounds, Lemma 4.2, and $\mathfrak{h}(s, t)^2 \leq 2|s - t|$ give

$$(4.33) \quad \begin{aligned} \mathfrak{h}(\hat{p}_{\text{raw}}, p), \mathfrak{h}(\hat{q}_{\text{raw}}, q) &\leq C\sqrt{\log(C/\eta)/n} + C\sqrt{\gamma} \\ &\leq c'\delta/\sqrt{N}. \end{aligned}$$

The constant c' can be made as small as required by the preceding choices of c_* and the sample multiplier. Sort the estimates and return $\hat{E} = \hat{p}P_{w_f} + \hat{q}(I - P_{w_f})$. Equations (7.9) and (4.5) make the spectral loss at most $\delta/4$; choose c_* so that (4.31) contributes at most $\delta/4$. Table 1 now gives the stated risk.

Finally we check the budget for every record. The product branch uses exactly $6n_0 + 8n$ calls. The entangled branch uses at most

$$(4.34) \quad 6n_0 + 4 \sum_{m=1,2,4,\dots,H} mB(\kappa, \epsilon_m) + 4m_*B(\zeta_*, \eta/8) + 2n.$$

This is a pathwise upper bound: the sum includes even the stages not reached after the first rejection. Its dyadic term is $O(N \log(C/\eta))$ by (4.25), including the ceilings in (4.19). For every returned $1 \leq m_* \leq N$, even on a failed record,

$$4m_*B(\zeta_*, \eta/8) \leq 4N + 16c_*^{-2}N\delta^{-2} \log(64/\eta).$$

Here $m_*\zeta_*^{-2} = c_*^{-2}N\delta^{-2}$ exactly. Since $N \geq 1$, $\delta \leq \delta_0$, and $\eta \leq 1/8$, both branches satisfy (4.2). The prescribed fallbacks always leave a unit direction and a legal effect, and every block has at most $H \leq N$ inputs. This proves the upper bound with the same worst-record convention as the converse.

For the converse consider the scalar effects

$$(4.35) \quad E_0 = \frac{1}{2}I, \quad E_1 = (\frac{1}{2} + \Delta)I, \quad \Delta = 256\delta/\sqrt{N}.$$

Since $\delta \leq 1/4096$, both parameters lie in $[1/2, 9/16]$. The finite Bernoulli lower bound in Lemma 7.2 yields

$$(4.36) \quad \begin{aligned} d_N(\mathcal{M}_{E_0}, \mathcal{M}_{E_1}) &= B_N(1/2, 1/2 + \Delta) \\ &\geq \frac{1}{64} \min\{1, \Delta\sqrt{N/(5/4)}\} > 2\delta. \end{aligned}$$

For these devices every call gives a fresh Bernoulli bit independent of the input. On an entangled input, each conditional reference block is the same reduced state multiplied by the scalar outcome probability. Thus any adaptive learning record, including its final classical estimate, is a common stochastic processing of M independent Bernoulli bits and parameter-independent resources. Publicly stopped records may be padded. The relative entropy before processing is at most

$$(4.37) \quad MD(\text{Bern}(1/2) \parallel \text{Bern}(1/2 + \Delta)) = -\frac{M}{2} \log(1 - 4\Delta^2) \leq 3M\Delta^2.$$

The two radius- δ success balls are disjoint by (4.36). Testing whether the estimated effect belongs to the first ball therefore gives two errors at most η . For classical laws P, Q , Jensen's inequality

gives root fidelity at least $e^{-D(P\|Q)/2}$. The trace-fidelity inequality then implies that the sum of testing errors is at least $\frac{1}{2}e^{-D(P\|Q)}$. Consequently

$$2\eta \geq \frac{1}{2}e^{-3M\Delta^2}, \quad M \geq \frac{N}{3 \cdot 256^2 \delta^2} \log \frac{1}{4\eta}.$$

For $\eta \leq 1/8$ this proves (4.3). $\square$

Corollary 4.6 (A learned joint description). *For rational δ in the range of Theorem 4.1, the learner may return one word of the rational joint code, with the same risk and call-count bounds. Its payload length is*

$$(4.38) \quad 2 \log_2 N + \log_2 \log(N+2) + 4 \log_2(1/\delta) + O(1), \quad 0 < \delta \leq \delta_0.$$

The decoding centre is a legal rational effect. This is a statement about device calls and payload length; classical arithmetic, workspace, and state-preparation costs are separate resources.

Proof. Use the learner with loss budget $\delta/2$. Its output is a finite computable expression determined by a finite record. In particular, the empirical frequencies are rational, and the guarded phase roots and normalized frames specify algebraic directions. Choose a positive rational $\varepsilon \leq \delta/(100N)$ and rational approximations (b_0, x_0) to its bias and Bloch vector with

$$|b_0 - \hat{b}| + \|x_0 - \hat{x}\| \leq \varepsilon.$$

Set $(b_1, x_1) = (b_0, x_0)/(1 + 2\varepsilon)$. Then

$$|b_1| + \|x_1\| \leq \frac{1 + \varepsilon}{1 + 2\varepsilon} \leq 1,$$

and the exact one-use formula (7.6) bounds the distance from the unrounded hypothesis by 3ε . The N -use hybrid inequality adds at most $3N\varepsilon$. Apply Theorem 8.2 to this rational estimated effect with public accuracy $\delta/2$. Its certified radius is $\delta/4$, so the total loss is less than δ . The capacity estimate in that theorem, at a constant rescaling of accuracy, gives (4.38). This postprocessing uses no additional device calls and receives no exact description of the unknown target. $\square$

The loss in Theorem 4.1 is a future adaptive distance. One-use measurement tomography is studied in [5, 29]; combining the stated one-use accuracy bounds only with the hybrid inequality gives a quadratic dependence on the future horizon. The argument here uses finite-use geometry and data-dependent entangled blocks to obtain the linear dependence in (4.2), with a matching scalar converse. Learning a stored programme from which a measurement is later retrieved, as in [30], has a different output resource. The present theorem gives a classical estimate over the entire ordered binary qubit body. Section 5 uses this common qubit procedure after a separate calibration to obtain the corresponding learning law on the full matrix effect body.

5. COMMON LEARNING IN ARBITRARY INPUT DIMENSION

The two-dimensional learner extends to the full effect body through a calibration step. The calibration must resolve the two support endpoints simultaneously. A bound in operator norm alone does not provide this information: after compression, a poorly aligned subspace can introduce noise at a scale larger than N^{-1} . The procedure below learns a basis in which this additional noise is bounded by the finite-horizon cutoff.

Throughout this section, $\mathcal{M}_E$ is the ordered binary measurement associated with $E \in \mathfrak{E}_d$. Training may use entangled input blocks and classical feedback, but it has no access to the measurement environment. Its output is a classical description of a legal effect. The future horizon N and the number of training calls remain separate resources.

Theorem 5.1 (Common learning on the matrix effect body). *There are absolute constants $c, C, \delta_0 > 0$ with the following property. For every $d, N \geq 1$, $0 < \delta \leq \delta_0$, and $0 < \eta \leq 1/8$, one common procedure returns a legal effect $\hat{E} \in \mathfrak{E}_d$ such that*

$$(5.1) \quad \sup_{E \in \mathfrak{E}_d} \mathbb{P}_E \{d_N(\mathcal{M}_E, \mathcal{M}_{\hat{E}}) > \delta\} \leq \eta.$$

On every record its number of calls satisfies

$$(5.2) \quad M \leq Cd^4 N \delta^{-2} \log(d/\eta),$$

and every entangled input block has length at most N . Conversely, any common procedure satisfying (5.1) with at most M calls on every record satisfies

$$(5.3) \quad M \geq cN \delta^{-2} \log(1/\eta).$$

The converse permits arbitrary reference-assisted adaptive training and bounded public stopping. In particular, for each fixed d , the minimax number of calls is $\Theta_d(N \delta^{-2} \log(1/\eta))$ on the complete closed effect body. The displayed dependence on d in the upper bound is not asserted to be optimal.

We first give the calibration, including its finite budgets. All matrix orders below are Loewner orders. The use of a basis selected from earlier observations is always followed by fresh samples.

Lemma 5.2 (Simultaneous endpoint calibration). *Let $\tau = N^{-1}$ and $0 < \epsilon \leq 1/8$. There is a common product-input procedure returning $A \in \mathfrak{E}_d$ with at most $Cd^4 N \log(d/\epsilon)$ calls on every record. With probability at least $1 - \epsilon$, it satisfies*

$$(5.4) \quad \frac{1}{2}(A + \tau I) \preceq E + \tau I \preceq 2(A + \tau I),$$

$$(5.5) \quad \frac{1}{2}(I - A + \tau I) \preceq I - E + \tau I \preceq 2(I - A + \tau I),$$

$$(5.6) \quad \|E - A\|_{\text{op}} \leq \sqrt{\tau}.$$

The input states depend only on the public parameters and the preceding record; no spectral data about E are supplied.

Proof. Set $J = \lceil \log_2 N \rceil$, $H = 2^J$, and $t_j = 2^{-j}$ for $0 \leq j \leq J$. Thus $N \leq H < 2N$ unless $N = 1$, and $\tau/2 < t_J \leq \tau$. Start from $A_0 = I/2$. If $J = 0$, return A_0 : all three conclusions hold without samples.

We use

$$h_d = d + 4 \binom{d}{2} = 2d^2 - d, \quad K_d = (320d)^2, \quad \alpha = 1/16.$$

At stage $j \geq 1$, assign failure allowance and sample size

$$(5.7) \quad \epsilon_j = \frac{\epsilon 2^j}{4H}, \quad L_j = \log \frac{2h_d}{\epsilon_j}, \quad n_j = \left\lceil K_d t_j^{-1} L_j \right\rceil.$$

Choose an orthonormal eigenbasis $e_1, \dots, e_d$ of the current legal estimate A_{j-1} . The choice is deterministic and measurable, including at multiple eigenvalues. More explicitly, fix a public input basis. All calibration matrices have algebraic entries in that basis: the sample frequencies are rational, and spectral projections, clipping at rational endpoints, and normalization preserve algebraic entries. Order the distinct eigenvalues, project the public basis onto each eigenspace, and apply Gram–Schmidt in that fixed order, discarding exactly zero vectors. Exact algebraic arithmetic therefore supplies a finite choice at every stage and requires no oracle for arbitrary real spectral data. Make n_j independent calls on each of the h_d states

$$(5.8) \quad e_i, \quad \frac{e_i + e_j}{\sqrt{2}}, \quad \frac{e_i - e_j}{\sqrt{2}}, \quad \frac{e_i + ie_j}{\sqrt{2}}, \quad \frac{e_i - ie_j}{\sqrt{2}} \quad (i < j).$$

Here and below a unit vector denotes its pure input state. Estimate each diagonal entry of E by its empirical outcome frequency. Estimate the real part of each off-diagonal entry by half the frequency difference between the positive and negative real superpositions. Its imaginary part is half the frequency on $(e_i - ie_j)/\sqrt{2}$ minus half the frequency on $(e_i + ie_j)/\sqrt{2}$. These formulas define a Hermitian matrix B_j .

To define the procedure on every record, clip the eigenvalues of B_j to $[-\alpha t_j, 1 + \alpha t_j]$, obtaining $\tilde{B}_j$, and set

$$(5.9) \quad A_j = \frac{\tilde{B}_j + \alpha t_j I}{1 + 2\alpha t_j}.$$

The returned matrix is always a legal effect. In particular, a failed concentration event cannot affect the subsequent call budget or require an additional experiment.

We prove the invariant

$$(5.10) \quad \frac{1}{2}(A_j + t_j I) \preceq E + t_j I \preceq 2(A_j + t_j I), \quad \frac{1}{2}(I - A_j + t_j I) \preceq I - E + t_j I \preceq 2(I - A_j + t_j I)$$

on the successive good events. It holds for $j = 0$ because $0 \preceq E \preceq I$.

Fix a preceding good record and abbreviate $t = t_j$, $A = A_{j-1}$, $B = B_j$, $D = B - E$, and $q = L_j/n_j$. All expectations and probabilities in the next calculation are conditional on that record. Bernstein's inequality and a union bound over the h_d groups show, with failure at most $2h_d e^{-L_j} = \epsilon_j$, that every empirical frequency has error at most

$$\sqrt{2p(1-p)q} + 2q/3.$$

For a pair of opposite superposition states, the sum of the outcome probabilities is $E_{ii} + E_{jj}$. Cauchy–Schwarz in each half-difference therefore gives the following two simultaneous bounds, also valid for $i = j$ after enlarging the constants:

$$(5.11) \quad |D_{ij}| \leq \sqrt{2(E_{ii} + E_{jj})q} + 2q, \quad |D_{ij}| \leq \sqrt{2(2 - E_{ii} - E_{jj})q} + 2q.$$

For an off-diagonal complex entry, apply the real estimate to the two quadratures and take their Euclidean norm. The factor $\sqrt{2}$ in (5.11) includes this operation.

Write a_i for the eigenvalues of A . The preceding invariant, whose regularization is $2t$, gives

$$E_{ii} \leq 2a_i + 2t, \quad 1 - E_{ii} \leq 2(1 - a_i) + 2t, \quad q \leq t/K_d.$$

Since

$$t(a_i + a_j + 2t) \leq 2(a_i + t)(a_j + t), \quad t \leq \sqrt{(a_i + t)(a_j + t)},$$

the first bound in (5.11) yields

$$(5.12) \quad \frac{|D_{ij}|}{\sqrt{(a_i + t)(a_j + t)}} \leq \frac{2\sqrt{2}}{\sqrt{K_d}} + \frac{2}{K_d} \leq \frac{5}{\sqrt{K_d}}.$$

Replacing a_i by $1 - a_i$ gives the same estimate for the complement. The Hilbert–Schmidt norm bounds the operator norm, so our choice of K_d implies

$$(5.13) \quad \begin{aligned} -\frac{1}{64}(A + tI) &\preceq D \preceq \frac{1}{64}(A + tI), \\ -\frac{1}{64}(I - A + tI) &\preceq D \preceq \frac{1}{64}(I - A + tI). \end{aligned}$$

The preceding invariant and $t_{j-1} = 2t$ also imply

$$A + tI \preceq 4(E + tI), \quad I - A + tI \preceq 4(I - E + tI).$$

Consequently

$$(5.14) \quad -\alpha(E + tI) \preceq D \preceq \alpha(E + tI), \quad -\alpha(I - E + tI) \preceq D \preceq \alpha(I - E + tI).$$

These inequalities place the spectrum of B inside $[-\alpha t, 1 + \alpha t]$. Thus the clipping in (5.9) does nothing on this event. On it,

$$A_j - E = \frac{D + \alpha t(I - 2E)}{1 + 2\alpha t}.$$

Since $-I \preceq I - 2E \preceq I$, equations (5.14) imply

$$\begin{aligned} -2\alpha(E + tI) &\preceq A_j - E \preceq 2\alpha(E + tI), \\ -2\alpha(I - E + tI) &\preceq A_j - E \preceq 2\alpha(I - E + tI). \end{aligned}$$

Adding the corresponding regularized endpoint gives factors $1 - 2\alpha = 7/8$ and $1 + 2\alpha = 9/8$. This proves the weaker factor-two invariant (5.10) for stage j .

The unweighted entry estimate also gives the operator-norm bound needed later. Because $E_{ii} + E_{jj} \leq 2$, we have

$$\|D\|_{\text{op}} \leq 4d\sqrt{t/K_d} = \sqrt{t}/80.$$

It follows that

$$(5.15) \quad \|A_j - E\|_{\text{op}} \leq \sqrt{t}/80 + \alpha t \leq \sqrt{t}.$$

At the final stage, adding $(\tau - t_J)I$ preserves the factor-two orders, and (5.15) gives (5.6).

The conditional failure allowances sum to less than $\epsilon/2$, since $\sum_{j=1}^J 2^j < 2H$. This proves the asserted probability bound. Finally, the deterministic call count is $h_d \sum_j n_j$. Writing $2^j = H/2^k$ yields

$$\sum_{j=1}^J 2^j L_j \leq H \sum_{k=0}^{J-1} 2^{-k} \left(\log \frac{8h_d}{\epsilon} + k \log 2 \right) \leq CH \log(d/\epsilon).$$

The ceiling terms contribute at most $h_d J \leq h_d H$. As $h_d K_d \leq Cd^4$ and $H < 2N$ when $N > 1$, the stated bound holds on every record. In particular, the number of dyadic stages does not introduce a factor $\log N$. $\square$

Lemma 5.3 (Calibrated variance and compression). *Suppose $A, E \in \mathfrak{E}_d$ satisfy (5.4)–(5.6). Set $W_G = G(I - G) + \tau I/2$. Then*

$$(5.16) \quad \frac{1}{12}W_A \preceq W_E \preceq 12W_A.$$

If P is any orthogonal projection commuting with A and $C = PEP|_{P\mathbb{C}^d}$, then, on $P\mathbb{C}^d$,

$$(5.17) \quad \frac{1}{12}PW_AP \preceq W_C \preceq 14PW_AP.$$

All constants are independent of the spectrum, its multiplicities, and the rank of P .

Proof. For a legal effect G , define the harmonic mean without a factor of two by

$$R_G = ((G + \tau I)^{-1} + (I - G + \tau I)^{-1})^{-1}.$$

The two factors commute because both are functions of G , so

$$(5.18) \quad R_G = \frac{G(I - G) + (\tau + \tau^2)I}{1 + 2\tau}, \quad \frac{1}{3}(G(I - G) + \tau I) \preceq R_G \preceq G(I - G) + \tau I.$$

The inequalities use only $0 < \tau \leq 1$. Inversion reverses Loewner order. Apply it to both endpoint comparisons, add, and invert again to get $R_E \preceq R_A$ with factors two. Equation (5.18) therefore compares $E(I - E) + \tau I$ and $A(I - A) + \tau I$ with factors six. Replacing the regularization τI by $\tau I/2$ proves (5.16).

The compression identity is exact:

$$(5.19) \quad C(I - C) = PE(I - E)P + PE(I - P)EP.$$

Because P commutes with A , $PE(I - P) = P(E - A)(I - P)$. Thus

$$0 \preceq PE(I - P)EP \preceq \|E - A\|_{\text{op}}^2 P \preceq \tau P.$$

The first term in (5.19), after adding $\tau P/2$, is $PW_E P$. Apply (5.16) and use $PW_A P \succeq \tau P/2$ to absorb the final τP with a factor two. This gives (5.17). $\square$

Proof of Theorem 5.1. For $d = 1$, Bernoulli sampling and Lemma 4.2 give the upper bound directly. Assume $d \geq 2$ and put $s = \binom{d}{2}$. Apply Lemma 5.2 with failure allowance $\eta/2$. Conditional on its good event, fix an orthonormal eigenbasis of the returned A , with eigenvalues a_i , and write

$$w_i = a_i(1 - a_i) + \tau/2.$$

For each $i < j$, let P_{ij} project onto $\text{span}(e_i, e_j)$ and let C_{ij} be the corresponding compression of E . An input qubit, including its entanglement with other input qubits or a retained reference, can be embedded isometrically into this two-dimensional subspace. The resulting observed channel is exactly $\mathcal{M}_{C_{ij}}$.

Run the common qubit learner of Theorem 4.1 on each compression, using fresh records, accuracy

$$(5.20) \quad \varepsilon = c_* \delta / d,$$

and failure allowance $\eta/(2s)$. The constant $c_* > 0$ will be chosen below. This returns legal 2×2 effects $\hat{C}_{ij}$. Every guarantee is conditional on the calibration record and is uniform over the resulting compression. A union bound therefore shows that, with total failure at most η , calibration succeeds and

$$(5.21) \quad d_N(\mathcal{M}_{C_{ij}}, \mathcal{M}_{\hat{C}_{ij}}) \leq \varepsilon \quad \text{for every } i < j.$$

We record explicitly how an operational error controls an endpoint form. Theorem 2.1 in dimension two and Lemma 2.5 imply that there are absolute $a_0, A_0 > 0$ such that, for legal $C, D \in \mathfrak{E}_2$,

$$(5.22) \quad d_N(\mathcal{M}_C, \mathcal{M}_D) \leq u \leq a_0 \implies g_{N,C}(D - C) \leq A_0 u.$$

Indeed, for $u < 1/16384$ the lower metric bound forces $Q_N(C, D) \leq 16384u$; decreasing a_0 makes this at most $1/4$, where the local comparison applies. Choose c_* small enough that $\varepsilon \leq a_0$ and lies within the qubit learner's accuracy range.

For a Hermitian matrix X on $\text{span}(e_i, e_j)$ put

$$\gamma_{ij}(X)^2 = N \left(\frac{|X_{ii}|^2}{2w_i} + \frac{|X_{jj}|^2}{2w_j} + \frac{2|X_{ij}|^2}{w_i + w_j} \right).$$

Equation (5.17), applied to left plus right multiplication and then inverted, gives

$$\gamma_{ij}(X) \leq \sqrt{14} g_{N,C_{ij}}(X).$$

Hence (5.21) and (5.22) imply

$$(5.23) \quad \gamma_{ij}(\hat{C}_{ij} - C_{ij}) \leq \sqrt{14} A_0 \varepsilon.$$

Assemble a Hermitian matrix B by taking each off-diagonal entry from its corresponding estimate $\hat{C}_{ij}$. For each diagonal entry choose once and for all one incident pair and use that pair's estimate. No consistency among different estimates of the same diagonal entry is assumed. In the diagonal coordinates of A , the form $g_{N,A}$ is the sum of the nonnegative coordinate terms displayed above. Each coordinate used in $B - E$ occurs in at least one of the pair forms, while unused duplicate diagonal terms can be discarded. Therefore

$$(5.24) \quad g_{N,A}(B - E)^2 \leq \sum_{i < j} \gamma_{ij}(\hat{C}_{ij} - C_{ij})^2 \leq 14s A_0^2 \varepsilon^2.$$

The assembled matrix need not be legal. Define the estimate by minimizing its distance to the legal effect body in the known positive quadratic norm $g_{N,A}$. The minimum is attained and is unique. An exact minimizer F satisfies

$$g_{N,A}(F - E) \leq 2g_{N,A}(B - E),$$

because E is a feasible competitor. To obtain a finite procedure with a rational legal output, it suffices instead to choose $F \in \mathfrak{E}_d$ with rational entries such that

$$(5.25) \quad g_{N,A}(F - B) \leq \inf_{G \in \mathfrak{E}_d} g_{N,A}(G - B) + \varepsilon.$$

This changes the bound only to

$$(5.26) \quad g_{N,A}(F - E) \leq 2\sqrt{14s}A_0\varepsilon + \varepsilon \leq C_0d\varepsilon$$

for an absolute C_0 .

For completeness, (5.25) is an effective finite selection, not a query to the unknown effect. Since $W_A \succeq \tau I/2$, $g_{N,A}(X) \leq N\|X\|_{\text{HS}}$. A finite rational grid covering $\mathfrak{E}_d$ in Hilbert–Schmidt norm to $\varepsilon/(4N)$ therefore covers it in $g_{N,A}$ to $\varepsilon/4$. Such a legal grid is obtained by rounding after the inward move $G \mapsto (1 - 2r)G + rI$: entrywise rounding with Hilbert–Schmidt error at most $r/2$ preserves positivity of both outcomes, and the total rounding displacement is at most $C_d r$. Concretely, choose positive rational $r \leq \varepsilon/[4N(\sqrt{d} + 1)]$ and a rational entry mesh at most $r/(2d)$. The inward displacement is at most $r\sqrt{d}$, and the rounding displacement is at most $r/2$, so their sum is at most $\varepsilon/(4N)$. The grid and its rational entries are defined in the fixed public input basis, independently of the random eigenbasis used for calibration. Enumerate this bounded grid, retain legal matrices using exact positivity tests, and evaluate its finitely many objective values to tolerance $\varepsilon/4$. Selecting the smallest computed value proves (5.25) with slack. The calibration matrices are algebraic, and the qubit subroutine’s outputs are finite computable expressions in its record, so these evaluations require no oracle for the unknown effect. Only the observed matrices, the public parameters, and finite classical calculations are used. Arithmetic and state preparation are not included in the device-call bound.

Finally, (5.16) gives

$$g_{N,E}(F - E) \leq \sqrt{12}g_{N,A}(F - E) \leq \sqrt{12}C_0d\varepsilon.$$

Choose the absolute c_* in (5.20) so that this is at most $\delta/(8\sqrt{2})$, and reduce δ_0 if necessary so that it is also at most $1/4$. The endpoint-to-midpoint comparison in Lemma 2.5 and the upper bound in Theorem 2.1 then give

$$d_N(\mathcal{M}_E, \mathcal{M}_F) \leq 8Q_N(E, F) \leq 8\sqrt{2}g_{N,E}(F - E) \leq \delta.$$

This proves the risk bound.

The calibration costs at most $Cd^4N \log(d/\eta)$ calls. There are $s \leq d^2/2$ qubit learners, each with accuracy $c_*\delta/d$ and confidence $\eta/(2s)$. Their combined call count is at most $Cd^4N\delta^{-2} \log(d/\eta)$ by Theorem 4.1. All budgets hold on every record, all compressed entangled blocks have length at most N , and postprocessing uses no additional calls.

The confidence and resource allocation can be recorded as follows; the constants in the call bounds are absolute.

Stage	Failure allowance	Calls on every record
Endpoint calibration	$\eta/2$	$Cd^4N \log(d/\eta)$
Each of $s = \binom{d}{2}$ pair learners	$\eta/(2s)$	$CN(d/\delta)^2 \log(d/\eta)$
Legal rational selection	0	0

The conditional guarantees of the pair learners allow their allowances to be added after calibration, despite the dependence of the selected basis on its record.

For the lower bound take

$$E_0 = \frac{1}{2}I_d, \quad E_1 = (\frac{1}{2} + \Delta)I_d, \quad \Delta = 256\delta/\sqrt{N}.$$

Choose $\delta_0 \leq 1/4096$. The Bernoulli separation in (4.36) is unchanged by the input dimension, and the two effects are separated by more than 2δ under d_N . Every use of either scalar device produces a fresh Bernoulli bit independent of the input. Its conditional reference state is the same reduced state multiplied by that scalar outcome probability. Thus an arbitrary adaptive training record with at most M calls, padded if necessary, is a common processing of M independent Bernoulli bits and parameter-independent resources. Its relative entropy is at most $3M\Delta^2$ by (4.37). The two disjoint success balls turn the estimated effect into a test with both errors at most η . The same fidelity inequality used in the proof of Theorem 4.1 gives

$$2\eta \geq \frac{1}{2}e^{-3M\Delta^2}, \quad M \geq \frac{N}{3 \cdot 256^2 \delta^2} \log \frac{1}{4\eta}.$$

For $\eta \leq 1/8$ this is (5.3). $\square$

The common learner now has the same dimension and boundary scope as the metric and entropy theorems. Its calibration is a sequence of product experiments, while the compression learners provide the enhanced angular resolution when an estimated two-dimensional subspace is close to opposite support endpoints. No pair of candidate hypotheses is supplied to either part of the procedure.

6. A FINITE RATIONAL CODE IN EVERY DIMENSION

The volume argument can be made into a single deterministic code whose decoder reconstructs its dictionary from the public parameters. The construction is finite and exact. Its purpose is to settle the constructive description problem; the enumeration below can be very large.

Theorem 6.1 (Deterministic rational matrix code). *Let $d, N \geq 1$ be integers and let $0 < \delta \leq 1$ be rational. There is a deterministic finite procedure, using rational arithmetic, that constructs an ordered list*

$$\mathcal{C}_{d,N,\delta} = (C_0, \dots, C_{M-1}) \subset \mathfrak{E}_d$$

of Gaussian-rational effects with the following properties. Every $E \in \mathfrak{E}_d$ has an assigned index $j(E)$ satisfying

$$(6.1) \quad d_N(\mathcal{M}_E, \mathcal{M}_{C_{j(E)}}) \leq \frac{11}{16}\delta.$$

For fixed d ,

$$(6.2) \quad M \leq C_d N^{d^2/2} [\log(N+2)]^{\lfloor d/2 \rfloor} \delta^{-d^2},$$

uniformly in N and δ . The encoder on a supplied Gaussian-rational effect and the decoder on its index both terminate using exact arithmetic. They require no matrix-dependent advice or preassigned list of centres. With (d, N, δ) public, the payload length is $\lceil \log_2 M \rceil$, and for $0 < \delta \leq \delta_d$ it is

$$(6.3) \quad \frac{d^2}{2} \log_2 N + \lfloor d/2 \rfloor \log_2 \log(N+2) + d^2 \log_2(1/\delta) + O_d(1).$$

Here δ_d may be the small-error constant in Theorem 3.1. The procedure makes no assertion of polynomial running time in the payload length.

Proof. Set

$$(6.4) \quad K = \left\lceil \frac{32dN}{\delta} \right\rceil, \quad s = \frac{2d}{K}, \quad \eta = \frac{\delta}{16}.$$

Fix the order of independent real coordinates to be the d diagonal entries, followed by the real and imaginary parts of G_{ij} for $i < j$ in lexicographic order. Let $\mathcal{G}_{d,K}$ consist of the legal Hermitian matrices whose diagonal coordinates belong to $\{0, 1/K, \dots, 1\}$ and whose remaining coordinates belong to $\{-1, -1 + 1/K, \dots, 1\}$. List these matrices in the induced lexicographic order. This list is finite, and it contains zero.

Traverse the list once. Retain its first element. Retain a subsequent element G exactly when

$$(6.5) \quad Q_N(G, C)^2 > \eta^2 \quad \text{for every previously retained element } C.$$

The retained elements, in their order of appearance, form $\mathcal{C}_{d,N,\delta}$. Every grid element either is retained or has a retained predecessor C with $Q_N(G, C) \leq \eta$. No triangle inequality for Q_N is used.

We first verify that the whole procedure is exact. Legality of a Gaussian-rational Hermitian matrix is decidable by exact Schur elimination, applied to G and $I - G$. A negative pivot excludes positive semidefiniteness. A zero pivot in a positive semidefinite matrix forces the corresponding row and column of the remaining matrix to vanish; otherwise the test rejects. A positive pivot reduces the question to the exact Schur complement. These rules terminate after at most d pivots, including on singular strata. For a legal rational pair, its midpoint A and variance $A(I - A)$ are Gaussian rational. The equation

$$A(I - A)X + XA(I - A) + N^{-1}X = G - C$$

has an invertible Gaussian-rational coefficient matrix. Solving it gives $Q_N(G, C)^2 = N \operatorname{tr}((G - C)X) \in \mathbb{Q}_{\geq 0}$. Consequently (6.5) is an exact rational comparison. It requires neither spectral coordinates nor a numerical eigenvalue threshold.

To encode E , first form the inward displacement

$$E_s = (1 - 2s)E + sI.$$

Since $s \leq 1/16$, its spectrum lies in $[s, 1 - s]$. Round each independent real coordinate of E_s to its nearest multiple of $1/K$, choosing the larger multiple at a tie, and call the resulting Hermitian matrix $G(E)$. A diagonal entry changes by at most $1/(2K)$ and the modulus of an off-diagonal entry changes by at most $\sqrt{2}/(2K)$. Thus

$$\|G(E) - E_s\|_{\text{op}} \leq \frac{d}{K}.$$

In particular,

$$(6.6) \quad \frac{d}{K}I \preceq G(E) \preceq \left(1 - \frac{d}{K}\right)I.$$

Every coordinate lies in the prescribed grid range, and hence $G(E) \in \mathcal{G}_{d,K}$. Moreover,

$$(6.7) \quad \|E - G(E)\|_{\text{op}} \leq s + \frac{d}{K} = \frac{3d}{K} \leq \frac{3\delta}{32N}.$$

Choose the first retained centre with $Q_N(G(E), C_j)^2 \leq \eta^2$. Such a centre exists by the completed greedy traversal. The hybrid inequality and Theorem 2.1 yield

$$\begin{aligned} d_N(\mathcal{M}_E, \mathcal{M}_{C_j}) &\leq 2N\|E - G(E)\|_{\text{op}} + 8Q_N(G(E), C_j) \\ &\leq \frac{3\delta}{16} + \frac{\delta}{2} = \frac{11\delta}{16}. \end{aligned}$$

This proves (6.1). If E is Gaussian rational, the prescribed rounding and every subsequent operation are exact finite rational calculations.

For two distinct retained centres, Theorem 2.1 gives

$$(6.8) \quad d_N(\mathcal{M}_C, \mathcal{M}_{C'}) \geq \frac{1}{8192d} \min\{1, Q_N(C, C')\} > \frac{\delta}{131072d},$$

where $\delta \leq 1$ was used. Let a_d, r_d be the constants of Lemma 3.2, and put

$$b_d = \min \left\{ r_d, \frac{1}{393216d} \right\}, \quad r = b_d \delta.$$

The operational balls of radius r about distinct retained centres are disjoint, by (6.8) and the triangle inequality for d_N . Their measures are each at least $a_d(b_d \delta)^{d^2}$. Hence

$$Ma_d(b_d \delta)^{d^2} \leq \mu_N(\mathfrak{E}_d).$$

Proposition 3.3 proves (6.2).

Both parties reconstruct this same finite list from (d, N, δ) . The encoder transmits the binary representation of j in $\lceil \log_2 M \rceil$ bits. The decoder reconstructs the list and returns C_j for $j < M$; every unused word of the same length is assigned the legal effect $I/2$. The codeword contains no additional centre coordinates. Equation (6.2) gives the upper bound in (6.3); the covering lower bound of Theorem 3.1 gives the matching lower bound, since this list is a radius- δ cover by (6.1). The description of the fixed reconstruction algorithm contributes only a constant. As usual, if the parameters are not public, their explicitly chosen header representation must be charged separately. $\square$

Real input and finite precision. The covering statement concerns the entire real effect body. To interpret its encoder computationally on a real input, a representation of that input must be supplied. A certified rational approximation of each independent real coordinate to error at most $1/(8K)$ is sufficient. Apply the inward displacement to this approximation and then use the same exact nearest-grid rounding. Relative to the true E_s , each independent coordinate then has error at most $5/(8K)$; each complex off-diagonal error is at most $5\sqrt{2}/(8K) < 1/K$. Therefore (6.6)–(6.7) and the full error bound remain valid. The exact supplied-matrix implementation takes Gaussian-rational input; no physical-device learning operation is implicit in that interface.

Computational scope. The ambient grid contains at most

$$(6.9) \quad (K+1)^d (2K+1)^{d(d-1)}$$

candidate coordinate tuples. The literal algorithm performs exact legality tests and, for each legal tuple, comparisons with the retained list. Retaining the list uses $O(Md^2)$ rational entries, and the decoder repeats the same construction. These are finite bounds, but their dependence on the requested horizon and precision prevents an efficiency assertion in the payload length. The exact kernel `matrix_codec.py` implements this specified enumeration, greedy test, rounding, and replay. A resource cutoff is reported as an incomplete construction and never as a codebook. Regression checks of small complete cases and of matrix kernels certify those computations; they do not assert that a full large-dimensional dictionary has been enumerated.

Corollary 6.2 (Learning into an optimal-order description). *For every fixed $d \geq 1$, every integer $N \geq 1$, every rational $0 < \delta \leq \min\{1, \delta_0, \delta_d\}$, and every $0 < \eta \leq 1/8$, there is one learner, common to all $E \in \mathfrak{E}_d$, which returns a fixed-length word whose deterministic public decoder produces a legal effect C with*

$$\sup_{E \in \mathfrak{E}_d} \mathbb{P}_E\{d_N(\mathcal{M}_E, \mathcal{M}_C) > \delta\} \leq \eta.$$

The training budget on every record is at most $Cd^4N\delta^{-2}\log(d/\eta)$, and each entangled block has at most N calls. The word has length

$$(6.10) \quad \frac{d^2}{2} \log_2 N + \lfloor d/2 \rfloor \log_2 \log(N+2) + d^2 \log_2(1/\delta) + O_d(1).$$

For fixed d , both the training order $N\delta^{-2}\log(1/\eta)$ and the payload order in (6.10) are optimal. Payload optimality is among fixed-length words with a fixed public decoder into legal memoryless ordered binary effects. Encoding after learning uses no additional calls to the unknown device.

Proof. Apply Theorem 5.1 with accuracy $\delta/2$. It returns a legal effect F with rational entries in the public input basis. Encode F using Theorem 6.1 with accuracy $\delta/2$. On the learning success event, the triangle inequality gives

$$d_N(\mathcal{M}_E, \mathcal{M}_C) \leq \frac{\delta}{2} + \frac{11\delta}{32} = \frac{27\delta}{32} < \delta.$$

The second operation is a deterministic computation on the recorded estimate. It uses no device calls and does not increase the failure probability. Replacing δ by $\delta/2$ changes the training constant and adds only $O_d(1)$ to the payload length.

The scalar subfamily converse in Theorem 5.1 applies to this learning task as well. For the payload converse, a B -bit word and fixed decoder have at most 2^B possible decoded effects. Every E has a strictly positive success probability, so at least one of these effects lies within distance δ of E . The decoded family is therefore a radius- δ cover of $\mathfrak{E}_d$. The lower bound in Theorem 3.1 gives (6.10) with the opposite inequality up to $O_d(1)$. $\square$

7. THE FULL BINARY EFFECT BODY

The spectral boundary of a biased measurement has two distinct endpoints. A vanishing eigenvalue of one effect need not give a coherent angular scale: the other outcome can still destroy coherence. We therefore retain both endpoint coordinates and determine their joint finite-use geometry.

Let

$$(7.1) \quad \mathcal{E} = \left\{ (a, x) \in \mathbb{R} \times \mathbb{R}^3 : 0 \leq a - |x| \leq a + |x| \leq 2 \right\}, \quad E_{a,x} = \frac{1}{2}(aI + x \cdot \sigma).$$

We identify a point (a, x) of $\mathcal{E}$ with its effect $E_{a,x}$. Write $\mathcal{M}_{a,x}$ for the measurement with ordered effects $E_{a,x}, I - E_{a,x}$ and no residual quantum output. Its maximum and minimum eigenvalues, spectral gap, and bias are

$$p = \frac{1}{2}(a + |x|), \quad q = \frac{1}{2}(a - |x|), \quad r = p - q = |x|, \quad b = p + q - 1.$$

Thus $0 \leq q \leq p \leq 1$ and $|b| + r \leq 1$. When $r > 0$ put $u = x/r$, so that $E_{a,x} = pP_u + q(I - P_u)$, where $P_u = (I + u \cdot \sigma)/2$. Directions are not identified under outcome exchange.

For an integer $N \geq 1$ set $\tau = N^{-1}$ and define

$$(7.2) \quad T_N(s, t) = |s - t| \sqrt{\frac{N}{\max\{s(1-s), t(1-t)\} + \tau}}, \quad 0 \leq s, t \leq 1,$$

$$V(p, q) = p(1-p) + q(1-q) = \frac{1}{2}(1 - b^2 - r^2),$$

$$(7.3) \quad K_N(p, q) = r \sqrt{\frac{N}{V(p, q) + \tau}}.$$

For two effects with data (p, q, u) and (p', q', v) let $h \in [0, \pi]$ be the angle between u, v when both spectral gaps are positive. If either gap is zero, set the following angular term equal to zero, without choosing a direction at the repeated eigenvalue:

$$(7.4) \quad H_N(E, F) = T_N(p, p') + T_N(q, q') + \min\{K_N(p, q), K_N(p', q')\} h.$$

Theorem 7.1 (Finite-use geometry of binary qubit effects). *For every $E, F \in \mathcal{E}$ and $N \geq 1$,*

$$(7.5) \quad \frac{1}{8192} \min\{1, H_N(E, F)\} \leq d_N(\mathcal{M}_E, \mathcal{M}_F) \leq \min\{2, 3H_N(E, F)\}.$$

The distance permits all common reference-assisted adaptive testers with at most N calls, including feedback and bounded public stopping, and uses the unhalved trace norm. The constants are independent of both spectra, both directions, and the horizon. In particular,

$$(7.6) \quad d_1(\mathcal{M}_{a,x}, \mathcal{M}_{a',x'}) = |a - a'| + |x - x'|.$$

The lower comparison is witnessed by pair-dependent nonadaptive tests. It does not require a common estimator of all four target coordinates.

The denominator in (7.3) involves the two outcome variances. Replacing it by the distance to the nearest spectral face would give an incorrect angular scale on a face such as $q = 0$, $0 < p < 1$. The proof below supplies both a scalar-overlap dilation upper bound and explicit finite-distance lower witnesses.

7.1. The spectral endpoints. Put

$$B_N(s, t) = \|\text{Bern}(s)^{\otimes N} - \text{Bern}(t)^{\otimes N}\|_1.$$

Lemma 7.2 (A Bernoulli estimate at both support endpoints). *For $s, t \in [0, 1]$,*

$$(7.7) \quad \frac{1}{64} \min\{1, T_N(s, t)\} \leq B_N(s, t) \leq \min\{2, 3T_N(s, t)\}.$$

For a fixed one of the two parameters, B_N increases as the other parameter moves away from it on either side.

Proof. Assume $s < t$, put $\Delta = t - s$, and write $v = \max\{s(1-s), t(1-t)\}$. The identical pair is immediate. A coupling of individual bits gives $B_N \leq 2N\Delta$. The root fidelity $F = \sqrt{st} + \sqrt{(1-s)(1-t)}$ satisfies

$$\begin{aligned} 1 - F &= \frac{\Delta^2}{2} \left(\frac{1}{(\sqrt{s} + \sqrt{t})^2} + \frac{1}{(\sqrt{1-s} + \sqrt{1-t})^2} \right) \\ &\leq \frac{\Delta^2(1+\Delta)}{2t(1-s)} \leq \frac{\Delta^2}{2v}, \end{aligned}$$

whenever $v > 0$. For the last inequality, both endpoint variances are at most $t(1-s)/(1+\Delta)$: after cancellation the required inequalities are $s(1+\Delta) \leq t$ and $(1-t)(1+\Delta) \leq 1-s$. They follow respectively from $\Delta(1-s) \geq 0$ and $t\Delta \geq 0$. The product-fidelity inequality gives $B_N \leq 2\sqrt{N}\Delta/\sqrt{v}$. For $z = Nv$, the inequality $\min\{2, 2/\sqrt{z}\}\sqrt{z} + 1 \leq 2\sqrt{2} < 3$ combines the two upper bounds. At $v = 0$ use the coupling bound.

For the lower bound, complement both bits if necessary to have $s + t \leq 1$. If $t > 3/4$, then $\Delta \geq 2t - 1 > 1/2$ and $B_N \geq B_1 = 2\Delta > 1$, which suffices. Otherwise $0 < t \leq 3/4$. Set

$$k = \min \left\{ N, \max \left\{ 1, \left\lfloor \frac{1}{2t} \right\rfloor \right\} \right\}, \quad m = \lfloor N/k \rfloor.$$

Then $k \geq \frac{1}{4} \min\{N, t^{-1}\}$, $m \geq N/(2k)$, and $(k-1)t \leq 1/2$. The event of no success in a block of k trials has probabilities $e_s = (1-s)^k$ and $e_t = (1-t)^k$, with

$$e_s - e_t \geq k\Delta(1-t)^{k-1} \geq k\Delta/2.$$

There is a common stochastic processing of this binary event whose success probability is $d + ce$, where

$$c = \frac{1}{\max\{e_s + e_t, 2 - e_s - e_t\}} \geq \frac{1}{2}, \quad d = \frac{1}{2}(1 - c(e_s + e_t)).$$

Indeed $d \geq 0$, $d+c \leq 1$, and the processed pair has parameters $(1 \pm c(e_s - e_t))/2$. Apply Lemma A.3 to m independent blocks and discard the unused calls. Contraction under this processing gives

$$B_N(s, t) \geq \frac{1}{16\sqrt{2}} \min\{1, \Delta\sqrt{N \min(N, t^{-1})}\}.$$

Since $v \geq t(1-t) \geq t/4$, $T_N(s, t) \leq 2\Delta\sqrt{N \min(N, t^{-1})}$. This proves the stated lower constant.

For monotonicity, the likelihood ratio of the two binomial count laws is monotone in the count. For a fixed larger parameter their total variation is the maximum, over upper count intervals, of the larger-parameter probability minus the smaller-parameter probability. The latter probabilities decrease when the smaller parameter decreases. Fixing the smaller parameter gives the analogous statement with lower count intervals. Endpoint cases follow by continuity of the finite sums. $\square$

Lemma 7.3 (Spectral projection without cancellation). *For arbitrary directions and spectra,*

$$(7.8) \quad B_N(p, p') \leq d_N(\mathcal{M}_E, \mathcal{M}_F), \quad B_N(q, q') \leq d_N(\mathcal{M}_E, \mathcal{M}_F).$$

If the two directions agree, then

$$(7.9) \quad d_N(\mathcal{M}_{p,q,u}, \mathcal{M}_{p',q',u}) \leq B_N(p, p') + B_N(q, q').$$

Consequently, with $d = d_N(\mathcal{M}_E, \mathcal{M}_F)$,

$$(7.10) \quad \begin{aligned} d_N(\mathcal{M}_{p,q,u}, \mathcal{M}_{p,q,v}) &\leq 3d, \\ d_N(\mathcal{M}_{p',q',u}, \mathcal{M}_{p',q',v}) &\leq 3d. \end{aligned}$$

Proof. Suppose first $p \geq p'$. Repeatedly input a maximum-eigenvalue eigenstate of E . Its success probabilities under the two hypotheses are p and some $t \leq p'$. The monotonicity in Lemma 7.2 gives $B_N(p, p') \leq B_N(p, t) \leq d$. If $p < p'$, interchange the hypotheses. For $q \leq q'$, use a minimum-eigenvalue eigenstate of E ; the other success probability is at least q' . Reverse the hypotheses if $q > q'$. These are separate admissible witnesses; no single input is asserted to witness both endpoint changes.

For aligned directions, first make the common projective measurement $\{P_u, I - P_u\}$ and, according to its outcome, generate a bit with success probability p or q . This implements exactly the measurement channel, including on inputs entangled with a reference: the reference blocks are the corresponding partial matrix elements weighted by the displayed success probabilities. Supply in advance N independent p -bits and N independent q -bits; at a call consume the next unused bit from the selected array. Every fixed adaptive tester is a common channel of these two arrays. The triangle inequality for tensor products therefore bounds its output distance by $B_N(p, p') + B_N(q, q')$. This proves (7.9) with arbitrary references, feedback, and stopping; all unqueried row bits are discarded. Combining it with (7.8) bounds every aligned spectral change by $2d$. The triangle inequality, aligning at u or at v , then proves (7.10). $\square$

7.2. Angular geometry with both eigenvalues retained.

Theorem 7.4 (Angular transition for a biased measurement). *For fixed $0 \leq q \leq p \leq 1$, $N \geq 1$, and directions at angle $h \in [0, \pi]$,*

$$(7.11) \quad \frac{1}{512} \min\{1, K_N(p, q)h\} \leq d_N(\mathcal{M}_{p,q,u}, \mathcal{M}_{p,q,v}) \leq \min\{2, \sqrt{2}K_N(p, q)h\}.$$

More precisely, put

$$A = \sqrt{p(1-p)} + \sqrt{q(1-q)}, \quad z = (p-q) \sin(h/2).$$

When $A > 0$,

$$(7.12) \quad d_N \leq \min \left\{ 2, 2Nz, 2\sqrt{1 - \left(\frac{A^2}{A^2 + z^2} \right)^N} \right\}.$$

When $A = 0$, the bound $\min\{2, 2Nz\}$ remains valid.

Proof. The one-use distance is $2z$: the effect difference is traceless with operator norm z , and the same reference trace-norm duality as in Lemma A.2 applies. Replacing the calls one at a time gives the adaptive bound $2Nz$.

We next construct a dilation with scalar cross overlap. Common unitary conjugation puts the two directions in one real meridian. Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad R_t = \exp(tJ), \quad t = h/2, \quad E = \text{diag}(p, q), \quad F = R_t E R_t^\top.$$

A measurement dilation is

$$W_E \psi = |+\rangle_C |+\rangle_L \sqrt{E} \psi + |-\rangle_C |-\rangle_L \sqrt{I - E} \psi.$$

The square-root vectors belong to a further two-dimensional environment. Tracing both environments leaves the ordered classical output and implements the measurement even with a reference. Put

$$f = \sqrt{pq} + \sqrt{(1-p)(1-q)}.$$

For any ϕ , apply the common rotation $R_{\phi-t}$ to the two-dimensional environment of the dilation W_F . The cross overlap is

$$\begin{aligned} W_E^\dagger (I \otimes R_{\phi-t}) W_F &= (\sqrt{E} R_\phi \sqrt{E} + \sqrt{I - E} R_\phi \sqrt{I - E}) R_{-t} \\ &= (\cos \phi I + f \sin \phi J) R_{-t}. \end{aligned}$$

Here the rotation acts as the identity on the outcome registers. When $f > 0$, choose

$$D = \sqrt{f^2 \cos^2 t + \sin^2 t}, \quad \cos \phi = \frac{f \cos t}{D}, \quad \sin \phi = \frac{\sin t}{D}.$$

Then the displayed overlap is cI , where $c = f/D \in [0, 1]$. This construction also covers $t = 0$ and $t = \pi/2$. To express it in the spectral coordinates, put $T = \sqrt{p(1-q)} + \sqrt{q(1-p)}$. Direct multiplication gives

$$fT = A, \quad (1 - f^2)T^2 = (p - q)^2.$$

If $A > 0$ then $f, T > 0$, and therefore

$$(7.13) \quad c^2 = \frac{f^2}{f^2 \cos^2 t + \sin^2 t} = \frac{A^2}{A^2 + (p - q)^2 \sin^2 t}.$$

Fix any common adaptive tester. Purify its initial state and all intermediate operations, and retain the measurement environments in this proof. The environments are never accessed by the tester. At each call the scalar identity for the cross overlap multiplies the inner product of the two purifications by c , independently of the current input and reference. The common intervening isometries preserve the inner product. A tester that stops early is padded to N calls on dummy inputs, with their outcomes ignored in its final output. Thus two purifications of that final output have inner product c^N . Trace-norm contraction and the pure-state formula give $d_N \leq 2\sqrt{1 - c^{2N}}$, proving (7.12). This is a pairwise dilation comparison; it does not require a processor for the entire effect body or an unknown-input environment-seizing procedure.

Since $V \leq A^2 \leq 2V$ and $1 - (1 + y)^{-N} \leq Ny$ for $y \geq 0$, the two nontrivial upper bounds imply

$$d_N \leq \min\{Nrh, \sqrt{N}rh/\sqrt{V}\} \leq \sqrt{2}K_N(p, q)h$$

when $V > 0$. The final inequality follows by putting $z = NV$ and using $\min\{1, z^{-1/2}\}\sqrt{z+1} \leq \sqrt{2}$. If $V = 0$, the only possibilities are scalar deterministic effects or a projective measurement, and the hybrid bound suffices.

For the lower bound, two explicit witnesses suffice. First suppose $r > 0$ and $h > 0$. Input a pure state in the direction $u - v$. Its success probabilities are

$$\frac{1 + b + r \sin(h/2)}{2}, \quad \frac{1 + b - r \sin(h/2)}{2}.$$

Their difference is $r \sin(h/2)$. The larger of their Bernoulli variances is at most $1 - b^2$: after complementing both output laws we may assume $a = 1 + b \leq 1$; both success probabilities lie in $[0, a]$, so their variances are at most $a \leq a(2 - a) = 1 - b^2$. Lemma 7.2, together with $\sin(h/2) \geq h/\pi$, now gives

$$(7.14) \quad d_N \geq \frac{1}{64\pi} \min \left\{ 1, rh \sqrt{\frac{N}{1 - b^2 + \tau}} \right\}.$$

The repeated pure input is a nonadaptive test.

For the second witness, choose a half-turn of the input qubit about an axis normal to the plane of u, v . It sends both Bloch directions to their negatives. With probability $1/2$ use the original measurement, and with probability $1/2$ apply this half-turn before the call and swap the two recorded outcomes afterwards. Discard the random choice. The resulting measurement has effects $(I \pm ru \cdot \sigma)/2$ or $(I \pm rv \cdot \sigma)/2$ under the respective hypotheses: the bias has been removed and the visibility is r . The same random preprocessing and postprocessing are used under both hypotheses and on every call. Any test for this unbiased pair is consequently an admissible test for the original pair. Theorem A.1 gives

$$(7.15) \quad d_N \geq \frac{1}{64} \min \left\{ 1, rh \sqrt{\frac{N}{1 - r^2 + \tau}} \right\}.$$

The witness is the finite entangled-block test of Lemma A.4, with this common processing. If the directions coincide or are antipodal, the required half-turn is chosen about any perpendicular axis.

Finally, feasibility $|b| + r \leq 1$ implies

$$\min\{1 - b^2, 1 - r^2\} \leq 2(1 - b^2 - r^2) = 4V.$$

For example, write $x = \max\{|b|, r\}$ and $y = \min\{|b|, r\}$; then $x \leq 1 - y$, $0 \leq y \leq 1/2$, and $x^2 + 2y^2 \leq (1 - y)^2 + 2y^2 \leq 1$, which is the asserted inequality. Hence the larger of the two coefficients in (7.14) and (7.15) is at least $K_N(p, q)/2$. Their stronger lower bound is at least $(128\pi)^{-1} \min\{1, K_N(p, q)h\}$, which implies (7.11). If $r = 0$ or $h = 0$, the two channels coincide and the assertion is immediate. $\square$

Proof of Theorem 7.1. For positive gaps use a path whose angular leg has the spectrum with the smaller angular coefficient. Lemma 7.3, followed by Lemma 7.2 and Theorem 7.4, gives

$$d_N(\mathcal{M}_E, \mathcal{M}_F) \leq 3T_N(p, p') + 3T_N(q, q') + \sqrt{2} \min\{K_N(p, q), K_N(p', q')\}h \leq 3H_N(E, F).$$

If one gap vanishes, that scalar effect may be aligned with the other direction at no cost, so the same estimate holds with angular term zero.

For the lower bound, put $A = T_N(p, p')$, $B = T_N(q, q')$, and $C = \min\{K_N(p, q), K_N(p', q')\}h$. The spectral projection lemma gives $d_N \geq 64^{-1} \min\{1, A\}$ and $d_N \geq 64^{-1} \min\{1, B\}$. Its angular projections and Theorem 7.4 give $d_N \geq 1536^{-1} \min\{1, C\}$. The largest of these three bounds is at least $4608^{-1} \min\{1, A + B + C\}$, which proves the weaker displayed constant $1/8192$. At a scalar effect only the two spectral bounds are needed. To check the assertion about lower witnesses, let d_N^{na} be the supremum over nonadaptive tests. The two spectral lower bounds hold with d_N^{na} , and the aligned spectral upper bound is consequently at most $2d_N^{\text{na}}$. Apply the triangle inequality to the output of a nonadaptive angular witness on the three-channel path. Its actual-pair leg is at

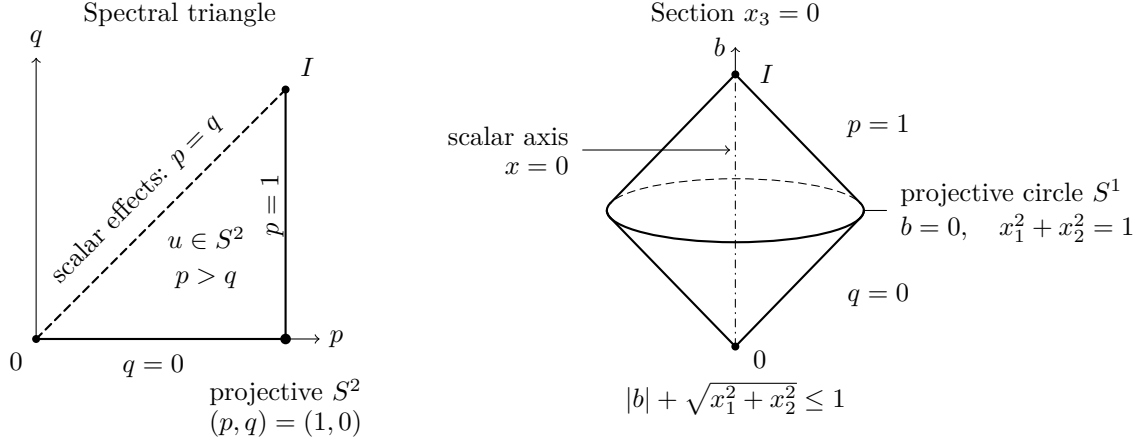


FIGURE 1. The four-dimensional effect body $|b| + |x| \leq 1$. Each point of the spectral triangle with $p > q$ carries a direction $u \in S^2$; on the scalar diagonal all directions represent the same effect. The boundary pieces $q = 0$ and $p = 1$ meet at the projective stratum $(p, q) = (1, 0)$, which carries the full sphere S^2 . The right panel is a projection of the three-dimensional section $x_3 = 0$; its equatorial circle is the intersection of that sphere with the section.

most d_N^{na} , and its aligned spectral leg is at most $2d_N^{\text{na}}$. The same lower comparison thus holds with d_N^{na} in place of d_N .

It remains to verify the one-use identity. For a reference-assisted input state ρ , the difference has output blocks B_ρ and $-B_\rho$, where $B_\rho = \text{tr}_A[(E - F) \otimes I]\rho$. Duality against Hermitian contractions gives $\|B_\rho\|_1 \leq \|E - F\|_{\text{op}}$. An extremal eigenstate of $E - F$ attains equality. Thus $d_1 = 2\|E - F\|_{\text{op}}$; the two eigenvalues of $E - F$ are $((a - a') \pm |x - x'|)/2$. This proves (7.6). $\square$

Remark 7.5 (Boundary faces and repeated eigenvalues). On $p = q = t$, the measurement is the input-independent coin $\text{Bern}(t)$ and all directions represent the same effect. Its exact finite-use distance along this scalar interval is $B_N(t, t')$. On the face $q = 0$, $0 < p < 1$, the angular coefficient is

$$K_N(p, 0) = p \sqrt{\frac{N}{p(1-p) + N^{-1}}}.$$

For a fixed p in this interval it has square-root-horizon order, despite the zero eigenvalue. At the projective edge $(p, q) = (1, 0)$ it equals N . At the two deterministic apices $p = q = 0$ and $p = q = 1$, the angular coefficient vanishes while the spectral moduli retain their finite-use support cutoff. These cases are part of the same theorem, without a spectral margin imposed on the target.

8. THE DESCRIPTION ENTROPY OF ALL BINARY QUBIT EFFECTS

Let

$$\mathfrak{E} = \{E = E^* \in \mathbb{C}^{2 \times 2} : 0 \leq E \leq I\}.$$

An effect E specifies the ordered measurement with effects $E, I - E$. Write $\mathcal{C}_N(\mathfrak{E}, \delta)$ for its covering number under d_N , allowing arbitrary legal memoryless centres of this interface, and write $\mathcal{C}_N^{\text{rat}}(\mathfrak{E}, \delta)$ when the centres have rational Cartesian coordinates. Both eigenvalues and the eigenspaces are charged. The spectral representation $E = pP_u + q(I - P_u)$, $p \geq q$, represents a scalar effect only once, independently of u .

Theorem 8.1 (The full effect body). *There are absolute constants $c, C, \delta_0 > 0$ such that, for every integer $N \geq 1$ and $0 < \delta \leq \delta_0$,*

$$(8.1) \quad cN^2 \log(N+2) \delta^{-4} \leq \mathcal{C}_N(\mathfrak{E}, \delta) \leq \mathcal{C}_N^{\text{rat}}(\mathfrak{E}, \delta) \leq CN^2 \log(N+2) \delta^{-4}.$$

Consequently the optimum fixed-length reusable payload is

$$(8.2) \quad 2 \log_2 N + \log_2 \log(N+2) + 4 \log_2(1/\delta) + O(1).$$

The constants are uniform over the whole closed effect body, including scalar, deterministic and projective effects. For rational accuracy input, rational target data admit an exact rational encoder attaining this order.

The logarithm records accumulation near the projective stratum, where both outcome probabilities can approach their support endpoints. We prove the upper bound by a finite rational construction and the lower bound by separated families of projective-corner boxes.

8.1. A rational spectral and angular code. For $N \geq 1$ and rational $0 < \delta \leq 1/4$, put

$$(8.3) \quad B = \left\lceil \sqrt{64N/\delta^2} \right\rceil, \quad s_i = \begin{cases} \frac{i^2}{B^2 + i^2}, & 0 \leq i \leq B, \\ 1 - \frac{(2B-i)^2}{B^2 + (2B-i)^2}, & B < i \leq 2B. \end{cases}$$

Thus $0 = s_0 < s_1 < \dots < s_{2B} = 1$. For $0 \leq j < i \leq 2B$ define

$$(8.4) \quad K_{ij}^2 = \frac{N(s_i - s_j)^2}{s_i(1-s_i) + s_j(1-s_j) + N^{-1}}, \quad A_{ij} = \max \left\{ 1, \left\lceil \sqrt{256K_{ij}^2/\delta^2} \right\rceil \right\}.$$

Use the six signed stereographic charts

$$(8.5) \quad u_a = \epsilon \frac{1 - |z|^2}{1 + |z|^2}, \quad u_b = \frac{2z_b}{1 + |z|^2} \quad (b \neq a), \quad a \in \{1, 2, 3\}, \quad \epsilon \in \{-1, 1\},$$

with $z \in \{-A_{ij}, \dots, A_{ij}\}^2/A_{ij}$ for the spectral pair (s_i, s_j) . Every chart word decodes to $s_i P_u + s_j(I - P_u)$. A diagonal pair $i = j$ has the single word $s_i I$. The exact number of words, including redundant chart words, is

$$(8.6) \quad L_{N,\delta} = (2B+1) + \sum_{0 \leq j < i \leq 2B} 6(2A_{ij} + 1)^2.$$

Theorem 8.2 (Exact rational realization). *The words in (8.6) form a radius- $\delta/2$ cover of $\mathfrak{E}$ by legal rational effects, and*

$$(8.7) \quad L_{N,\delta} \leq CN^2 \log(N+2) \delta^{-4}$$

for an absolute C . If

$$E = \frac{1}{2}((1+b)I + x \cdot \sigma), \quad b \in \mathbb{Q}, \quad x \in \mathbb{Q}^3, \quad |b| + |x| \leq 1,$$

then a covering word is computable by rational arithmetic and integer comparisons. Its payload is one fixed-length index in the set of (8.6).

Proof. Write $f(t) = t^2/(1+t^2)$, $0 \leq t \leq 1$. For a probability in $[0, 1/2]$, round its inverse coordinate t to the nearest multiple of $1/B$; for a probability in $[1/2, 1]$, use the reflected chart $1 - f(t)$. Choose the multiple of smaller magnitude at ties. This defines a nondecreasing quantizer on $[0, 1]$, so separately quantizing $p \geq q$ preserves their order.

The square-root vector associated with $f(t)$ is $(t, 1)/\sqrt{1+t^2}$; its derivative has norm $1/(1+t^2) \leq 1$. The reflected chart has the same bound. Hence the Euclidean square-root distance between each probability and its quantization is at most $1/(2B)$. If H is this distance, the product-fidelity

inequality gives unhalved distance at most $2\sqrt{N}H$ between the corresponding N -fold Bernoulli laws. The common two-coin programme in Lemma 7.3 therefore gives a spectral error at most

$$2\sqrt{N}/B \leq \delta/4.$$

This argument includes adaptive testers and entangled references.

If the quantized eigenvalues coincide, this completes the encoding. Otherwise choose a signed chart whose distinguished coordinate has maximal absolute value in the target direction. Its inverse coordinate belongs to $[-1, 1]^2$. Rounding its two coordinates to the grid of denominator A_{ij} gives angular error at most $2\sqrt{2}/A_{ij}$, since geodesic distance is bounded by the length of the mapped straight segment and the differential of stereographic projection has norm at most two. The angular upper bound in Theorem 7.4 gives error at most

$$\sqrt{2}K_{ij} \frac{2\sqrt{2}}{A_{ij}} \leq \delta/4.$$

The triangle inequality proves the covering assertion. Formula (8.5) preserves the unit sphere exactly, so every decoded effect is legal and rational.

We verify exact computability even when the input eigenvalues are irrational. Put $Q = |x|^2 \in \mathbb{Q}$. The eigenvalues are $(1 + b \pm \sqrt{Q})/2$. Each decision in the spectral rounding compares one of these numbers with a rational threshold $f((k + 1/2)/B)$ or its reflection. Such a comparison is the sign of a rational number plus or minus $\sqrt{Q}$; checking signs before squaring makes it an exact rational comparison. If $Q = 0$, the two quantized eigenvalues coincide. Otherwise choose the smallest index a maximizing $|x_a|$ and put $\epsilon = \text{sgn}(x_a)$. The inverse chart is

$$z_b = \frac{x_b}{\sqrt{Q} + |x_a|} \quad (b \neq a).$$

Its comparison with any rational rounding threshold again reduces to the sign of a rational number plus a rational multiple of $\sqrt{Q}$. The integers B, A_{ij} are computed by integer squaring and rational comparison. Thus no irrational normalization is required. The decoder uses only the displayed rational formulae.

It remains to count the words. Set $m_i = \min(i, 2B - i)$ and $U = B^2/N$. Directly from (8.3),

$$\frac{m_i^2}{4B^2} \leq s_i(1 - s_i) \leq \frac{m_i^2}{B^2}, \quad K_{ij}^2 \leq \frac{4NB^2}{m_i^2 + m_j^2 + U}.$$

Each value of m_i occurs at most twice. Since $(2A_{ij} + 1)^2 \leq C(1 + K_{ij}^2/\delta^2)$, we obtain

$$(8.8) \quad L_{N,\delta} \leq CB^2 + \frac{CNB^2}{\delta^2} \sum_{a,b=0}^B \frac{1}{a^2 + b^2 + U}.$$

Here $U \geq 1$. The part with $\max(a, b) \leq \sqrt{U}$ is bounded by an absolute constant. Each successive dyadic square ring contributes at most another absolute constant. There are at most $C \log(N + 2)$ such rings, because $B/\sqrt{U} = \sqrt{N}$. Consequently

$$\sum_{a,b=0}^B \frac{1}{a^2 + b^2 + U} \leq C \log(N + 2).$$

Finally $B^2 \leq CN/\delta^2$, so (8.8) proves (8.7). The support cutoff in the denominator is retained throughout the count, including the endpoint grid words. $\square$

8.2. Packing the projective corner.

Proof of Theorem 8.1. The upper bound follows from Theorem 8.2. For a real δ , choose a rational δ' with $\delta \leq \delta' \leq 2\delta$ and use its radius- $\delta'/2$ code; decrease δ_0 to at most $1/8$. For the lower bound first let $N \geq 16$. For every

$$t = 8^j/N \leq 1/16, \quad j \geq 0,$$

consider the effects with

$$(8.9) \quad p = 1 - \alpha, \quad q = \beta, \quad t \leq \alpha, \beta \leq 2t, \quad u = u(z), \quad z \in [-1/4, 1/4]^2,$$

where $u(z)$ is the chart in (8.5) with $a = 3, \epsilon = 1$. On this square, spherical distance is bounded above and below by absolute positive multiples of Euclidean distance in z . Throughout (8.9), the eigenvalue gap is at least $3/4$ and

$$p(1-p) \asymp q(1-q) \asymp t, \quad p(1-p) + q(1-q) + N^{-1} \asymp t.$$

The finite-distance comparison of Theorem 7.1 therefore supplies an absolute $c_1 > 0$ such that any two points in this box satisfy

$$(8.10) \quad d_N \geq c_1 \min \left\{ 1, \sqrt{N/t} |(\alpha, \beta, z) - (\alpha', \beta', z')|_\infty \right\}.$$

Choose an absolute sufficiently large A and then an absolute sufficiently small δ_0 . In each of the four coordinates take a grid of spacing $A\delta\sqrt{t/N}$. Since $Nt \geq 1$, this spacing is at most half the width of either spectral interval when $\delta \leq \delta_0$. The resulting points have pairwise distance at least 4δ by (8.10), and their number is at least

$$c \left(\frac{t}{\delta\sqrt{t/N}} \right)^2 \left(\frac{1}{\delta\sqrt{t/N}} \right)^2 = cN^2\delta^{-4}.$$

Points from different boxes are also uniformly separated. Indeed, if $t' \geq 8t$, then $\alpha' \geq t'$ and $\alpha \leq 2t \leq t'/4$. The spectral modulus of the corresponding upper eigenvalues is bounded below by an absolute positive multiple of $\sqrt{Nt'} \geq 1$. The spectral noncancellation bound in Theorem 7.1 gives $d_N \geq c_2 > 0$. Shrinking δ_0 ensures that this is at least 4δ . There are at least $c \log(N+2)$ permitted boxes. Their union is therefore a 4δ -packing of cardinality $cN^2 \log(N+2)\delta^{-4}$.

For $1 \leq N < 16$, use instead a fixed spectral rectangle $5/8 \leq p \leq 3/4$, $1/4 \leq q \leq 3/8$ and the same angular chart. Theorem 7.1 gives a uniform local Euclidean lower bound, and a grid of spacing $C\delta$ has at least $c\delta^{-4}$ points. The factor $N^2 \log(N+2)$ is bounded on this finite range. This proves the same lower estimate for all $N \geq 1$.

A radius- δ ball about any legal centre contains at most one point of a 4δ -packing, by the triangle inequality. Hence the packing proves the lower bound with the stated centre convention. Taking binary logarithms and rounding the index length proves (8.2). $\square$

8.3. The source of the logarithm. The spectral and angular weights also give the following integral form of the same accumulation. Put $v_p = p(1-p)$, $v_q = q(1-q)$ and $\tau = N^{-1}$. Then

$$(8.11) \quad \int_{0 \leq q \leq p \leq 1} \frac{N^2(p-q)^2 dp dq}{(v_p + v_q + \tau) \sqrt{(v_p + \tau)(v_q + \tau)}} \asymp N^2 \log(N+2).$$

To verify this, near $p = 1, q = 0$ put $\alpha = 1 - p$, $\beta = q$. After removing the factor N^2 , the integrand is comparable to

$$\frac{1}{(\alpha + \beta + \tau) \sqrt{(\alpha + \tau)(\beta + \tau)}}.$$

For $s = \alpha + \beta$, integration over $0 \leq \alpha \leq s$ is bounded above by an absolute constant times $1/(s + \tau)$. For $s \geq 2\tau$, restriction to $s/4 \leq \alpha \leq 3s/4$ gives the matching lower bound. Integration

in s proves logarithmic growth. Near the scalar endpoint $p = q = 0$, the factor $(p - q)^2$ is at most $(p + q)^2$, while $v_p + v_q \asymp p + q$; the remaining integral is uniformly bounded. Complementation gives the same conclusion near $p = q = 1$. Away from these endpoint neighbourhoods, $v_p + v_q$ is bounded below and the factors $v_p^{-1/2}, v_q^{-1/2}$ are integrable. Bounded N is absorbed into the constants. Thus the extra logarithm is contributed by the projective corner. The finite covering construction and packing above also include the collapsed scalar coordinates, where the spectral representation ceases to be a chart.

9. MEASUREMENT DISCRIMINATION, GEOMETRY, AND LEARNING

The present binary interface retains the ordered classical outcome and an arbitrary external reference, but no postmeasurement quantum system from the device. This fixes the comparison with the measurement-discrimination literature. The projective theorem of Puchała and collaborators [16, Corollary 1 and Theorem 2] gives an exact multiple-use formula with ancillary access. It is used in the retained observable-readout theorem. The restricted ancillary interface in Fiurášek–Mičuda [19] leads to a different comparison of adaptive and parallel probes.

Krawiec–Pawela–Puchała [20, Section 4] construct a pair of nine-outcome qutrit rank-one POVMs that admit perfect adaptive discrimination with two calls and no perfect discrimination by any finite parallel experiment. Datta–Biswas–Saha–Augusiak [21, Theorems 3–5] study entanglement-assisted single-use discrimination, including perfect constructions and a qubit entanglement advantage. Their qubit construction has three outcomes. These are direct antecedents for general POVM discrimination with a retained reference. They do not supply a full binary-effect family covering law. Our comparison (2.23) permits an adaptive advantage and controls its size uniformly in N for fixed input dimension.

The differential argument in Lemma 2.2 belongs to the established horizontal-gauge and channel-extension method [22, 7, 23, 24]. Yuan–Fung give the parallel path estimate [25, Section III, equation (26), PDF version 3]; Sieniawski–Demkowicz-Dobrzański integrate adaptive bounds [26, Section IV.A]. The binary Sylvester choice here gives the specific regularized variance $M(I - M)$, which can be integrated explicitly and matched by compressed nonadaptive tests. The inverse-Sylvester expressions of Bures geometry [27] and its spectral integration [28] are related methods. Our form acts on the effect displacement H with variance $V = M(I - M)$; it is not the ordinary Bures pullback under $M \mapsto V$, whose differential would be $H - MH - HM$. The operational-ball lemma and the nested endpoint integral are both required for Theorem 3.1.

Mele–Bittel [5, Theorem I.1] prove the fixed-confidence channel-learning bound $O(d_{\text{in}}d_{\text{out}}k\epsilon^{-2})$ for Kraus rank at most k and $0 < \epsilon < 1$. Their binary specialization [5, Lemmas III.7–III.8 and Corollary III.9] gives

$$\|\mathcal{M}_E - \mathcal{M}_F\|_{\diamond} = 2\|E - F\|_{\text{op}}, \quad M = 1024d^2\epsilon^{-2} + O(d\epsilon^{-2}\log(1/\eta)),$$

with $\|\hat{E} - E\|_{\text{op}} \leq \epsilon$ and probability at least $1 - \eta$, provided $\eta > 4e^{-4d^2}$. The channel-rank bound is $k = 2d$; no effect-rank assumption is imposed. Their channel calls prepare independent maximally entangled probes in parallel, followed by collective processing of the resulting Choi states. The learner has no measurement–environment access. For arbitrary confidence, [5, Remark III.4] gives the repetition bound $O(d^2\epsilon^{-2}\log(1/\eta))$. The dimension lower bound matches at fixed accuracy; it is distinct from their separate accuracy lower bound.

Zambrano–Ramos-Calderer–Kueng [29, Definition 1] use the worst-input one-use outcome total variation, denoted d_{op} . Their Theorem 2, equation (13), gives

$$M \geq \frac{8(d^2 + \epsilon(d^2 + 1)/6)}{\epsilon^2} \log \frac{2^{L+1}9^{2d}}{\eta}$$

as a sufficient sample size for error ϵ and confidence $1 - \eta$, using known probes from a global 2-design and classical least-squares reconstruction followed by projection onto the POVM body. Calls are nonadaptive and single-copy, without retained references or entanglement between calls. The result holds for arbitrary effect ranks and L outcomes. Their Theorem 9 gives the corresponding $\Omega((d^3 + d^2 L)\epsilon^{-2})$ lower bound at constant confidence within that access model. For $L = 2$, $d_{\text{op}}(E, F) = \|E - F\|_{\text{op}}$.

Thus both works already provide a common estimator of an unknown binary measurement. To compare their stated guarantees with the present loss, the hybrid inequality gives

$$d_N(\mathcal{M}_E, \mathcal{M}_{\hat{E}}) \leq 2N\|E - \hat{E}\|_{\text{op}}.$$

Substituting $\epsilon = \delta/(2N)$ into either cited binary guarantee gives a sufficient training budget $O_d(N^2\delta^{-2}\log(1/\eta))$. This substitution certifies a sufficient budget; it makes no converse assertion about those estimators under the future-use loss. Theorem 5.1 proves $\Theta_d(N\delta^{-2}\log(1/\eta))$ directly for d_N , uniformly over the complete binary matrix effect body for each fixed d , for $N \geq 1$, $0 < \delta \leq \delta_0$, and $0 < \eta \leq 1/8$. Its training procedure chooses entangled input blocks of length at most N using the observed record, and returns one legal classical estimate. The lower bound allows arbitrary retained references, adaptive controls, and bounded public stopping. The explicit upper bound is $Cd^4N\delta^{-2}\log(d/\eta)$; its dimension dependence is not asserted to be sharp. Calibration of both support endpoints and noise-adapted compression learning are what permit the statistical theorem to share the full matrix scope of the metric theorem. Multiscale phase estimation and robust phase recovery are established techniques [31, 32]. Our proof supplies its own small-cone argument and conditional confidence allocation; it does not invoke the numerical unwrapping threshold corrected by the erratum to [32]. Randomized signs remove the unknown bias exactly, and empirical amplitudes choose the entangled block length without a supplied noise parameter. The qubit subprocedure is common to all unknown qubit effects. The calibrated compression construction extends common learning to the full matrix body in every fixed dimension. The pair-dependent tests used in the matrix metric comparison and the supplied-target exact encoder retain their separate roles.

The deterministic rational construction in Theorem 6.1 adds a finite exact selection rule to the optimal-order covering theorem. Its exhaustive computation is accounted for separately from the payload; no efficient high-dimensional encoder or optimal arithmetic complexity is inferred.

All matrix entropy constants are for fixed dimension and small error. The proof includes every rank and spectral multiplicity of the binary effect body. Multi-outcome POVMs, residual quantum outputs, sharp growing-dimension minimax and entropy laws, and full submaximal-error matrix coverings require additional arguments. The inherited full-error preparation and fixed-readout results retain their own stated ranges.

APPENDIX A. A NOISE-UNIFORM READOUT TRANSITION

The dependence on the readout basis can change scale when noise approaches an extremal measurement. We determine that transition in a fixed one-dimensional family, without a positive spectral margin uniform in the noise. The proof is a finite-distance argument; no conclusion is inferred merely from a Fisher-information calculation.

Let σ_1, σ_2 be the first two Pauli matrices. For $\theta \in \mathbb{R}/(2\pi\mathbb{Z})$ and a public visibility $0 \leq \lambda \leq 1$, put

$$(A.1) \quad E_{\theta, \lambda}^{\pm} = \frac{1}{2}(I \pm \lambda(\cos \theta \sigma_1 + \sin \theta \sigma_2)), \quad \mathcal{M}_{\theta, \lambda}^{\pm}(X) = \text{tr}(E_{\theta, \lambda}^{\pm} X).$$

Both outcome labels are retained. The output quantum system is one-dimensional. In particular, θ and $\theta + \pi$ are not identified by an outcome relabelling. Let $h(\theta, \phi) \in [0, \pi]$ be circular distance

and set

$$(A.2) \quad \eta = 1 - \lambda^2, \quad K_{N,\lambda} = \lambda \sqrt{N \min\{N, \eta^{-1}\}}, \quad N \geq 1,$$

where $\eta^{-1} = +\infty$ at $\lambda = 1$. Thus $K_{N,1} = N$ and $K_{N,0} = 0$. The visibility may depend arbitrarily on N and the requested error; it is not a target coordinate charged in this theorem.

Write $\mathcal{C}_{N,\lambda}(\delta)$ for the least number of legal memoryless binary instruments of this interface whose d_N -balls of radius δ cover all targets in (A.1). Centres need not belong to the displayed family. Define $\mathcal{C}_{N,\lambda}^{\text{rat}}(\delta)$ by restricting centres to family members with rational $\cos \theta, \sin \theta$; rational visibility then gives rational effects. A constant in the following statement is independent of all three parameters N, λ, δ .

Theorem A.1 (Noise-uniform finite-use coding). *For all $N \geq 1$, $0 \leq \lambda \leq 1$, and all pairs of angles,*

$$(A.3) \quad \frac{1}{64} \min\{1, K_{N,\lambda} h(\theta, \phi)\} \leq d_N(\mathcal{M}_{\theta,\lambda}, \mathcal{M}_{\phi,\lambda}) \leq \min\{2, K_{N,\lambda} h(\theta, \phi)\}.$$

There are absolute $c, C > 0$ such that, for $0 < \delta \leq 2^{-10}$,

$$(A.4) \quad c \left(1 + \frac{K_{N,\lambda}}{\delta}\right) \leq \mathcal{C}_{N,\lambda}(\delta) \leq \mathcal{C}_{N,\lambda}^{\text{rat}}(\delta) \leq C \left(1 + \frac{K_{N,\lambda}}{\delta}\right).$$

Consequently the optimum fixed-length reusable payload is

$$(A.5) \quad \log_2 \left(1 + \frac{K_{N,\lambda}}{\delta}\right) + O(1).$$

The upper construction is an exact rational algorithm on rational visibility, tolerance and unit-circle input coordinates. At $\lambda = 0$ the family is a singleton and its optimum payload is zero.

The constants and the additive remainder are uniform up to the erased and projective endpoints. In contrast, they do not assert an optimal leading constant, a large-error coherent covering law, or a coding law in which an unknown visibility is also encoded.

A.1. A two-point programme with its noise dependence retained.

Lemma A.2 (An adaptive upper modulus). *Let $h = h(\theta, \phi)$ and $a = \lambda \sin(h/2)$. If $\eta > 0$, then*

$$(A.6) \quad d_N(\mathcal{M}_{\theta,\lambda}, \mathcal{M}_{\phi,\lambda}) \leq \min \left\{ 2, 2Na, 2\sqrt{1 - \left(\frac{\eta}{\eta + a^2}\right)^N} \right\}.$$

For $\eta = 0$ the bound $\min\{2, 2Na\}$ remains valid.

Proof. The difference of the two positive-outcome effects has operator norm a . On a bipartite input state, the two output blocks of the difference are B and $-B$, where $B = \text{tr}_A[(E_{\theta,\lambda}^+ - E_{\phi,\lambda}^+) \otimes I]\rho$. Duality against Hermitian contractions gives $\|B\|_1 \leq a$. An eigenstate of the effect difference attains equality. Hence $d_1 = 2a$, including reference assistance. Replacing the calls one at a time and using trace-norm contraction proves $d_N \leq 2Na$ for every common adaptive tester.

Choose angle representatives with difference h and let u, v be the unit vectors along their midpoint and its oriented perpendicular in the equatorial plane. The two Bloch vectors are $m \pm av$, where $m = \lambda \cos(h/2)u$. Put $b = (1 - \|m\|^2)^{1/2} = (\eta + a^2)^{1/2}$. The two Bloch vectors $m \pm bv$ have norm one, so they define two legal binary projective instruments, say $\mathcal{A}_+$ and $\mathcal{A}_-$. The target pair is exactly the two mixtures of these same instruments with programme laws

$$p_{\pm} = ((1 \pm t)/2, (1 \mp t)/2), \quad t = a/b.$$

This construction depends on the pair, which is permissible in a metric upper bound. It is not one fixed processor for the whole circle.

For a fixed tester, first supply N independent programme bits and, at each requested call, apply the indicated $\mathcal{A}_+$ or $\mathcal{A}_-$. All subsequent operations, including the reference, feedback and a stopping rule, are a common channel of these bits. Unused bits are discarded. The root fidelity of p_+, p_- is $\sqrt{1-t^2}$. Purifying the classical laws and using contraction gives

$$\|p_+^{\otimes N} - p_-^{\otimes N}\|_1 \leq 2\sqrt{1 - (1-t^2)^N}.$$

This also follows directly from the classical fidelity inequality. Substitution proves (A.6). Finally $1 - (1-t^2)^N \leq Nt^2$ and $a \leq \lambda h/2$ imply

$$d_N \leq \min\{N\lambda h, \sqrt{N/\eta} \lambda h, 2\}.$$

The limiting projective bound is supplied by the hybrid estimate, not by division by zero. $\square$

Classical simulation by common extremal channels and the effect of noise on the Heisenberg scale are established methods [7]. Environment/programme-state processing under adaptive use is also an established framework [13, 14, 15]. The point needed here is the exact two-point expression (A.6) and its uniform conversion to a family cover, rather than a new claim about the origin of those methods.

A.2. Finite entangled blocks and a binary product estimate.

Lemma A.3 (Symmetric Bernoulli products). *Let $P_\pm$ be Bernoulli laws with success probabilities $(1 \pm a)/2$, where $0 \leq a \leq 1$. For every integer $m \geq 1$,*

$$(A.7) \quad \|P_+^{\otimes m} - P_-^{\otimes m}\|_1 \geq \frac{1}{2} \min\{1, a\sqrt{m}\}.$$

Proof. Retain the largest odd number $r = 2j + 1 \leq m$ of coordinates, so $r \geq m/2$. For their majority event write $g_r(a) = P_+(\text{majority}) - P_-(\text{majority})$. Differentiating the finite binomial sum and cancelling consecutive terms gives

$$g'_r(a) = r \binom{2j}{j} 4^{-j} (1-a^2)^j.$$

For $j \geq 1$, the bound $\binom{2j}{j} 4^{-j} \geq 1/(2\sqrt{j})$ follows by induction: the ratio of consecutive left sides is $(2j+1)/(2j+2) \geq \sqrt{j/(j+1)}$. For $0 \leq a \leq r^{-1/2}$, Bernoulli's inequality gives $(1-a^2)^j \geq 1-j/r \geq 1/2$. Thus $g'_r(a) \geq \sqrt{r}/(2\sqrt{2})$ on that interval. The case $r = 1$ is immediate. Integration from zero and monotonicity yield $g_r(a) \geq (2\sqrt{2})^{-1} \min\{1, a\sqrt{r}\}$. Unhalved distance is at least twice this event difference; using $r \geq m/2$ proves the stated bound. $\square$

Lemma A.4 (A noise-uniform pairwise witness). *The lower inequality in (A.3) is witnessed by nonadaptive entangled blocks, each of at most N qubits, followed by classical parity and majority processing.*

Proof. A block of k inputs is prepared in $(|0\rangle^{\otimes k} + e^{i\alpha}|1\rangle^{\otimes k})/\sqrt{2}$. Multiplying the k output signs of (A.1) has mean $\lambda^k \cos(k\theta - \alpha)$. This identity follows by applying $\bigotimes_{i=1}^k (E^+ - E^-)$ to the two off-diagonal terms of the block density matrix. For the two hypotheses choose the one common phase $\alpha = k(\theta + \phi)/2 + \pi/2$. Their parity laws then have opposite biases of magnitude

$$(A.8) \quad a_k = \lambda^k |\sin(kh/2)|.$$

Repeating $m = \lfloor N/k \rfloor$ independent blocks and ignoring remaining calls is an admissible tester. Its unhalved output distance is at least $\frac{1}{2} \min\{1, \sqrt{m} a_k\}$ by Lemma A.3. The phase and block size may depend on the pair, but never on the unknown hypothesis or a private outcome.

Assume $\lambda > 0$ and put $T = \min\{N, \eta^{-1}\}$, $k_0 = \lfloor T \rfloor$. Then $T/2 \leq k_0 \leq T$. For every $1 \leq k \leq k_0$,

$$(A.9) \quad \lambda^k \geq \lambda/2.$$

Indeed, when $0 < \eta < 1$, the inequality $\log(1 - \eta) \geq -\eta/(1 - \eta)$ and $k - 1 \leq \eta^{-1} - 1$ give $(k - 1) \log \lambda \geq -1/2$. At $\eta = 0$ the assertion is immediate.

If $h \leq 1/k_0$, choose $k = k_0$. Since $\sin(kh/2) \geq kh/\pi$ and $m \geq N/(2k)$, we obtain

$$\sqrt{m} a_k \geq \frac{\lambda \sqrt{N k_0}}{2\pi \sqrt{2}} h \geq \frac{K_{N,\lambda}}{4\pi} h.$$

This gives a lower constant $1/(8\pi)$. If $k_0 = 1$, use $k = 1$ for all $h \leq \pi$ instead. Now $T < 2$ unless $T = 1$, and $\sqrt{N} \lambda \sin(h/2) \geq K_{N,\lambda} h/(\pi \sqrt{2})$; this is stronger than required.

It remains that $k_0 \geq 2$ and $h > 1/k_0$. Here $\lambda \geq 1/\sqrt{2}$. Choose $k = \max\{1, \lfloor 1/h \rfloor\}$. Then $k \leq k_0$ and $1/2 \leq kh \leq \pi$. Equations (A.8) and (A.9) give $a_k \geq 1/(4\pi \sqrt{2})$. Even one such block has distance at least $1/(8\pi \sqrt{2}) > 1/64$. The three cases prove the claim. For $\lambda = 0$ both sides of the desired inequality vanish. $\square$

A.3. Covering and exact rational realization.

Proof of Theorem A.1. The metric assertion follows from the preceding lemmas. For a lower cover choose points on the circle with circular separation at least $256\delta/K_{N,\lambda}$ whenever that quantity is at most π . An equally spaced packing has at least a fixed constant times $K_{N,\lambda}/\delta$ points. Every two points have d_N -distance at least 4δ , because $256\delta \leq 1/4$. No radius- δ ball, even about an arbitrary legal instrument, can contain two such points. If the required angular separation exceeds π , the trivial lower bound one suffices, with a smaller absolute constant. This proves the first inequality in (A.4).

For the upper bound use the two rational semicircle charts

$$(A.10) \quad z_s(t) = s \left(\frac{1 - t^2}{1 + t^2}, \frac{2t}{1 + t^2} \right), \quad s \in \{-1, 1\}, \quad -1 \leq t \leq 1.$$

Each chart has angular speed $2/(1 + t^2) \leq 2$. Round t to the nearest j/B , $-B \leq j \leq B$, with a fixed rule at ties. The circular error is at most $1/B$. Thus any integer $B \geq \max\{1, K_{N,\lambda}/\delta\}$ yields a cover with at most $4B + 2$ centres. For rational inputs, an entirely rational choice is

$$(A.11) \quad B = \max \left\{ 1, \min \left\{ \left\lceil \frac{4N\lambda}{\delta} \right\rceil, \left\lceil \sqrt{\frac{16N\lambda^2}{\eta\delta^2}} \right\rceil \right\} \right\},$$

omitting the second entry when $\eta = 0$. Integer squaring and rational comparison compute the ceiling square root exactly. Both entries are ceilings of the corresponding real bounds, so their minimum is $\lceil 4K_{N,\lambda}/\delta \rceil$. For $z = (x, y)$ on the rational unit circle choose $s = 1$ when $x \geq 0$ and $s = -1$ otherwise, and set $t = sy/(1 + sx)$. Its denominator is at least one, so it is rational and belongs to $[-1, 1]$.

The chart digit (s, j) determines a legal effect pair with eigenvalues $(1 \pm \lambda)/2$. No approximate positivity test, irrational normalization or search over real angles is needed. A fixed-length index over $4B + 2$ slots includes the chart sign. Its length gives (A.5). Redundant chart endpoints cause at most a constant overhead. At $\lambda = 0$ dispense with the chart and emit the single fair, input-erasing measurement. $\square$

Corollary A.5 (A continuous family of horizon exponents). *Fix $0 < \delta \leq 2^{-10}$. If $\lambda_N \rightarrow 1$ and $1 - \lambda_N^2 = N^{-\alpha+o(1)}$ for $\alpha \geq 0$, then*

$$\frac{\log_2 \mathcal{C}_{N,\lambda_N}(\delta)}{\log_2 N} \longrightarrow \frac{1 + \min\{\alpha, 1\}}{2}.$$

If $\lambda_N = N^{-\beta+o(1)} \rightarrow 0$, with $\beta \geq 0$, the limit is $\max\{1/2 - \beta, 0\}$. The projective and erased endpoints have coefficients one and zero, respectively.

Proof. Substitute the two regimes in (A.2) and use (A.4). The term $1 + K/\delta$ handles a vanishing or subpolynomial scale without taking the logarithm of zero. $\square$

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